



High-resolution palaeoenvironmental reconstruction at Zmajevac (Croatia) over the last three glacial/interglacial cycles

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ARTICLE INFO

Editor: Paul Hesse

Keywords:

Loess chronology
Environmental magnetism
Grain-size analysis
Clay minerals
Diffuse reflectance spectrometry

ABSTRACT

A high-resolution analysis of the Zmajevac loess-palaeosol sequence in East Croatia (Baranja) was performed with the purpose to get new insights into the palaeoenvironmental history during the last three glacial-interglacial cycles. The post-IR infrared stimulated luminescence signal of feldspar, stimulated at 290°C (thereafter pIRIR₂₉₀) was used to establish a reliable chronological framework and revealed intensive sedimentation during (at least) the last 300 kyr, which was five times interrupted by climate optima resulting in the formation of soils (now palaeosols). The environmental magnetic measurements support the obtained luminescence chronology. Further chronological time control was reached by age modelling based on tie points from variations in frequency dependence of magnetic susceptibility and benthic $\delta^{18}\text{O}$ variations from the LR04 stack of Lisiecki and Raymo (2005). The grain-size record provided information about the environmental conditions and dust accumulation. Additional palaeoenvironmental implications were provided by room temperature magnetic susceptibility and colourimetric analyses and revealed a continuously lower intensity of pedogenesis, coupled with constantly decreasing temperatures and precipitation from the oxygen isotope stage (OIS) 9 to the end of OIS 3. X-ray diffraction of the clay mineral fraction revealed chlorite, smectite and illitic material as dominant mineral phases. While clay minerals from loess are mostly detrital, smectite from palaeosols is partly of a pedogenic origin. The application of our multi-proxy approach, including relative and numerical dating techniques, provided new insights into a highly resolved terrestrial palaeoenvironmental record of Eastern Croatia.

1. Introduction

The climate of the last several million years has been influenced by orbital- and millennial-scale fluctuations which are archived in various sediments. The most intensively investigated terrestrial Quaternary sedimentary archives by far are loess-palaeosol sequences (LPSs), which are important deposits genetically linked to major river systems (Smalley et al., 2009) and the extent of the glaciations. Loess-palaeosol sequences may provide detailed data on terrestrial dust accumulation processes and climate changes (Frechen et al., 2003; Marković et al., 2015; Schaetzl et al., 2018). The loess deposits preserved in the Chinese Loess Plateau (CLP) in Central China exhibit a quasi-continuous sediment accumulation during glacial and interglacial cycles during the last

2.4 Ma (Heller and Liu, 1982) and provide a detailed record of climate change in the Asian interior (Kukla, 1987) while the last million years are also recorded in European loess-palaeosol sequences (Antoine et al., 2021; Basarin et al., 2014; Hlavatskyi and Bakhmutov, 2020; Laag et al., 2021a, 2021b; Marković et al., 2015; Schaetzl et al., 2018; Song et al., 2018; Sümegi et al., 2018).

The Danube is the second largest river of Europe and drains an area of ~800,000 km² (Sommerwerk et al., 2009) and along with its many tributaries, is also a major transport means of silt i.e. loess material (Smalley et al., 2009). The Danube loess region is an important geo-archive where numerous thick loess-palaeosol sequences have been investigated (Marković et al., 2015; Schaetzl et al., 2018) but often lack absolute and reliable chronologies, especially for the Middle Pleistocene.

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<https://doi.org/10.1016/j.palaeo.2021.110504>

Received 4 December 2020; Received in revised form 18 May 2021; Accepted 25 May 2021

Available online 29 May 2021

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A robust chronological framework is mandatory for palaeoenvironmental reconstructions. For that purpose, luminescence dating methods are applied, in particular, the post-IR infrared stimulated luminescence (pIRIR), which is nowadays the most popular dating approach when it comes to dating of older deposits (~ 300 ka; e.g. Fenn et al., 2020; Wacha et al., 2013). Due to the lack of adequate dating methods for Middle Pleistocene sediments, environmental magnetism is also helpful for placing chronological constraints (e.g. Evans and Heller, 2001; Heller and Liu, 1982 – Chinese loess; Basarin et al., 2014; Laag et al., 2021a, 2021b; Marković et al., 2011; Zeeden et al., 2016 – Danube loess).

Palaeoenvironmental information is obtained by numerous proxy data often in high-resolution which are then compared with global records (LR04 stack; Lisiecki and Raymo, 2005). High-resolution grain size analysis provides data about dust mobilisation, transport and deposition mechanisms (Újvári et al., 2016a; Vandenberghe, 2013) as well as post-depositional conditions related to the climate during and shortly after its deposition (e.g. Schulte and Lehmkuhl, 2018; Vandenberghe et al., 2018). Furthermore, grain size data can be used to qualitatively reconstruct wind intensities and directions (Bokhorst et al., 2011).

Another useful palaeoenvironmental proxy is the magnetic susceptibility (χ) and its frequency dependence ($\Delta\chi$), which is a direct result of the soil-forming processes and the neoformation of magnetic minerals during pedogenesis. In most loess areas of the Eurasian loess belt, both χ and $\Delta\chi$ are increased in buried soils (palaeosols) and relatively low in loess units (e.g. Maher, 1986). χ and $\Delta\chi$ depend strongly on the composition, relative abundance and grain size of different magnetic and paramagnetic minerals inside a bulk sample (e.g. Maher, 2011). The highest values are reached in samples containing the soft magnetic minerals, magnetite and maghemite, which already influence the bulk-samples' signal in amounts of less than 1 (wt)% (e.g. Maher et al., 2003). However, magnetic minerals important for environmental reconstructions like hematite and goethite are barely traceable by χ . The magnetic signal is influenced by the magnetic grain size and thus domain state as well, with single domain (SDs), pseudo-single domain (PSDs) and multi-domain (MDs; Liu et al., 2012) grains. Especially the ultrafine super-paramagnetic particles (with a diameter ranging up to 30 nm) are formed in the course of pedogenesis. The neoformation of these superparamagnetic (SP) magnetite/maghemite particles close to the unstable single domain (SP) to stable single domain (SD) grain sizes (30 nm) takes place during pedogenesis (favouring warm and humid conditions) (Heller and Liu, 1982; Peters and Dekkers, 2003). χ and $\Delta\chi$ are used as proxies for palaeoprecipitation (e.g. Hao and Guo, 2005; Heller et al., 1993; Liu et al., 1995; Maher et al., 1994). Here, we use $\Delta\chi$ for (1) discriminating between warm/moist interglacials and cold/dry glacials, and (2) to compare $\Delta\chi$ fluctuations with variations in other well-dated master records representing global climate oscillations such as the LR04 stack (Lisiecki and Raymo, 2005).

The combination of magnetic and colourimetric investigations is a valuable tool for unravelling and understanding the different mineralogy and their forming conditions which are a direct result of the environmental impact (e.g. Laag et al., 2021a; Lukić et al., 2014). Different magnetic minerals may record a variety of environmental conditions like precipitation, temperature, and their process-related oscillations in sub-millennial timeframes (e.g. maghemite – Gao et al., 2019). Hence, changes in relative abundances of minerals, as magnetite, maghemite, hematite and goethite are imprints of the palaeoenvironment onto a terrestrial geoarchive. Hematite and goethite cannot be quantified easily through room temperature susceptibility measurements, hence time-consuming as well as complicated measurements are necessary (e.g. low-temperature susceptibility measurements; Lagroix and Guyodo, 2017) to discriminate them. Diffuse reflectance spectrometry (DRS) and the first derivative bands of the backscattered spectra provide numerical indicators for the relative abundance of hematite and goethite (e.g. Wu et al., 2018).

Colour changes in loess are important indicators for both changing concentrations and compositions of clay minerals, organic matter and hematite, controlled by different environmental conditions in terms of precipitation, temperature, and seasonality (Ji et al., 2002). Whereas relative amounts of magnetite and maghemite express high χ values in a low concentration ($< 1\%$), hematite and goethite in loess can hardly be detected by fluctuations in χ . Investigations of soil-colours were performed during the last decades with Munsell-colour charts, while recent developments for colour-measurements by visible-DRS with spectrophotometers provide objective numerical values to detect small amounts of hematite and goethite inside of bulk-samples (Deaton and Balsam, 1991; Wu et al., 2018).

The clay minerals content and their formation are related to the parent material and source area, transport processes, the weathering regime of the sediment, as well as climate and chemical conditions during their formation (e.g. Taheri et al., 2020; Weaver, 1989). Under arid and cold climates, clay minerals remain relatively unmodified, while humid conditions can significantly alter them. For that reason, they are a powerful tool for interpreting weathering conditions and can be used as a proxy for palaeoclimate reconstructions (e.g. Kovács et al., 2013; Sheldon and Tabor, 2009).

The Danube loess belt preserves palaeoenvironmental records of continuous dust accumulation that extend over the last million years and is characterized by alternating loess formation during glacial times and pedogenesis during interglacial times (e.g. Bradák et al., 2019; Jordanova et al., 2007; Laag et al., 2021b; Marković et al., 2015; Song et al., 2018; Sümegi et al., 2018). These deposits are also the thickest due to the high sediment supply from the Danube and its tributaries and high preservation potential due to pre-existing basin topography (Fitzsimmons et al., 2012; Obrecht et al., 2019). For this reason, numerous multiproxy studies were performed in recent years to understand the significance of LPs and to unravel the data they archive especially related to palaeoclimate (e.g. Laag et al., 2021a; Lukić et al., 2014; Zeeden et al., 2016). Furthermore, the Danube and its tributaries were also the main dispersal corridor (Hauck et al., 2018) for early modern humans during the Upper Palaeolithic which resulted in many Early Upper Palaeolithic sites in the region (Chu, 2018). Detailed stratigraphy of the Danube loess is proposed by (Marković et al., 2015) based on numerous proxy data as well as absolute chronology (Murray et al., 2014; Schmidt et al., 2010). As for loess in eastern Croatia along the Danube, the stratigraphy is based on infrared stimulated luminescence (IRSL; Galović et al., 2009) and post-IR IRSL dating (Wacha et al., 2013; Wacha and Frechen, 2011). Loess along the Drava river, also a tributary of the Danube, was investigated by (Wacha et al., 2018a).

The Zmajevac loess-palaeosol section is probably the most detailed loess record exposed in Baranja (eastern Croatia) and can therefore be used to establish the loess stratigraphy for the area. Furthermore, it is a valuable terrestrial sedimentary archive of palaeoenvironmental changes for the Middle and Upper Pleistocene period. A previous chronological study of the Zmajevac loess-palaeosol sequence was performed by (Galović et al., 2009), where the stratigraphy of the exposed loess and palaeosol horizons was given accompanied with first IRSL dating results. Their ages however, are likely underestimated because these were not corrected for fading (Huntley and Lamothe, 2001; Wintle, 1973). For this study, the same location was excavated and sampled. The Zmajevac loess-palaeosol section was re-investigated through high-resolution sampling and by applying a multi-proxy methodology (high-resolution grain-size analysis, magnetic susceptibility measurements, clay mineral content, and diffuse reflectance spectrometry). Here, we present high-resolution grain-size records supplemented by χ and DRS data from the Zmajevac loess section and use the results to unravel the interaction between the palaeoclimate and individual proxies. Furthermore, a more detailed and extended chronological framework is presented by luminescence dating supported by correlative age models.



Fig. 1. Location map with indicated most relevant loess-palaeosol sections in the area mentioned in the text.

2. Geological setting and the Zmajevac loess section

The Zmajevac loess-palaeosol section (45.813° N, 18.820° E) is located in Baranja in NE Croatia, which is bordered by the Danube in the east, the Drava in the south and the Hungarian state border in the north-

northwest. Baranja is the SE margin of the Pannonian basin and it is mostly represented by thick sequences of Quaternary sediments which are genetically related to the complex river network and their response to Pleistocene climate changes (Fig. 1). Underlying the Quaternary deposits, Palaeozoic metamorphic rocks and Miocene and Pliocene sediments were

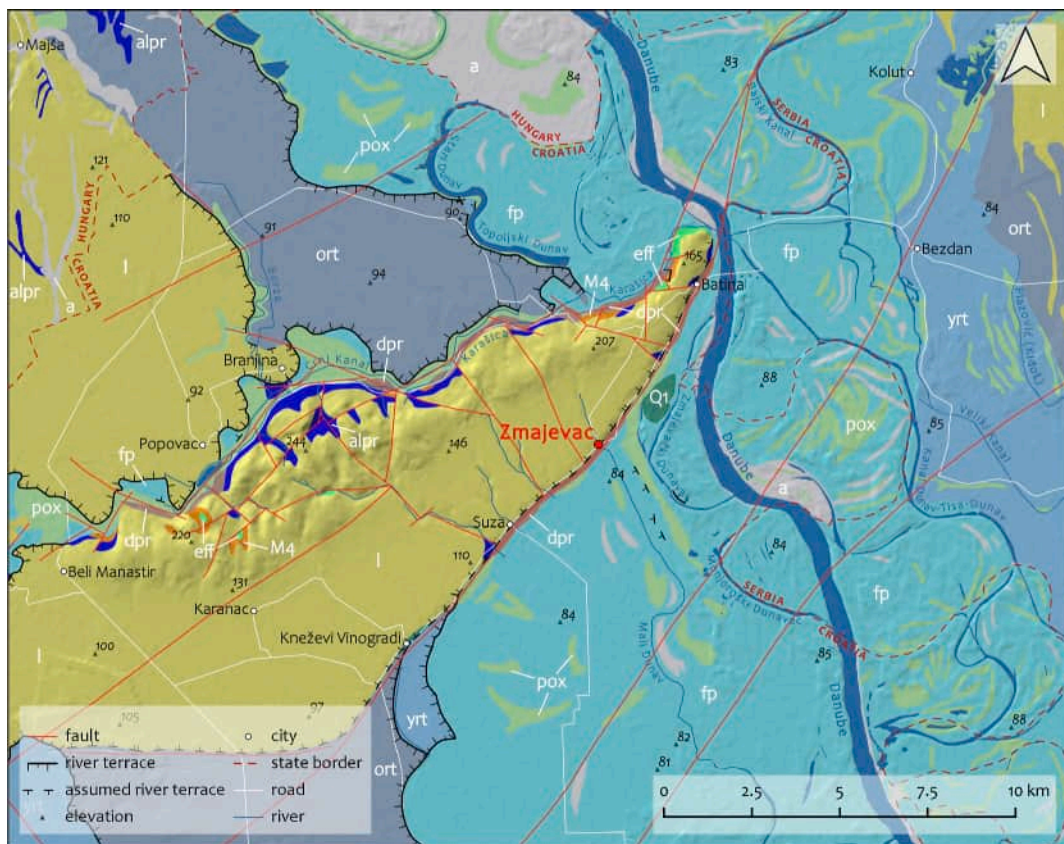


Fig. 2. Geological map of the investigated area (simplified after Gyalog L. and Sikhegyi F. 2005; Pikija et al., 1991a; Trifunović and Strajin, 1995). Legend: Holocene: a - Recent alluvial, beach, ait and bar deposits; yrt - Younger river terrace: Bars and floodplain deposits; dpr - Deluvial - proluvial deposits; pox - Pond and oxbow deposits; Pleistocene: fp - Floodplain deposits; alpr - Alluvial, lake and proluvial deposits; l - Loess and loess-like deposits; ort - Older river terrace: Oxbow, alluvial and loess; Q1 - Unclassified clastites (clay, silt, sand, gravel and peat); Miocene (Badenian): eff - Tuffs, volcanic breccias and basaltic andesite; M4 - Marl, sandy marl, limestones, breccia limestones, sand, sandstones, conglomerates, montmorillonite clay.

determined in deep boreholes, while on the surface mostly Badenian (Miocene) sediments and volcanic rocks can be seen underlying the loess (Pikija et al., 1991a, 1991b, 1995). Except for Bansko brdo (brdo = hill), Baranja is a lowland area, resulting from the alluvial interaction of the rivers Drava, Danube and Karašica (Figs. 1 & 2). The older river terraces are covered by Pleistocene loess and loess-like deposits forming a loess plateau. Bansko brdo is a tectonically uplifted area marked by reverse faults of opposing orientation, on the surface showing Badenian sediments and volcanic rocks covered by loess (Pikija et al., 1991a, 1991b). Baranja is influenced by active tectonics, as evidenced by the present morphostructure and main faults (Hećimović, 2005) some of them forming steep loess cliffs, as seen along the Zmajevac-Batina road (Fig. 2) where the investigated section is exposed (Fig. 3).

The Zmajevac loess-palaeosol section is situated at the outskirts of the Zmajevac settlement, about 6 km away from the Croatian-Serbian state border. It is situated at the fault zone of the Bansko Brdo tectonic unit (Pikija et al., 1991b), at the SE slope of the Bansko Brdo elevated morphostructure (Figs. 1 & 2). The investigated sequence is around 29 m thick. The section starts with an around 3 m thick layer of loess-like sediment, alluvial clayey to sandy silt (Fig. 3). This unit is rich in up to 1 mm Fe—Mn concretions, reddish-brown variegation and redox-imorphic features which imply to water saturation. It also shows traces of strong biological activity in the form of bioturbation, krotovinas and root remains (rhizoconcretions). Furthermore, it contains carbonate concretions up to 10 cm. This horizon represents a loess-like sediment probably deposited by aeolian activity in a waterlogged environment, i. e. on the flood plain. In the upper part of this horizon, about 50 cm of loess is present.

The lowest i.e. the oldest exposed palaeosol covering the lowermost loess (Fig. 3), is 250 cm thick and was described by (Galović et al., 2009) as a pedocomplex interrupted by several erosional discontinuities. Furthermore, this pedocomplex was described to be a soil-sediment that was exposed to pedogenesis after redeposition of the soil (Galović, 2014). The lower part of this pedocomplex is enriched with up to 0.5 cm large carbonate concretions, contains krotovinas and has a relatively higher sand content. The palaeosol has a reddish-brown colour. It is covered by a 4 m thick loess layer/horizon on top of which another pedocomplex is developed (Galović, 2014; Galović et al., 2009) which consists of two brown palaeosols intercalated by a 50 cm thick loess layer. The pedocomplex is covered by a laminated sand layer, around 230 cm thick, and shows a sharp and abrupt discontinuity with the layer below. The lower part of the loess covering the laminated sand layer has an increased amount of sand and fines upwards. This loess layer is covered by another brown palaeosol, about 190 cm thick, and shows bioturbation and krotovinas in its upper part. Covering the brown palaeosol is the uppermost, stratigraphically youngest loess horizon which is in its upper part intercalated by a weakly developed, up to 200 cm thick palaeosol. On the top of the section a modern rendzina soil is developed (Galović et al., 2009).

3. Methods

3.1. Sampling and sample preparation

During fieldwork in 2015, the previously investigated locations (Galović et al., 2009) were revisited and sampled for a more detailed

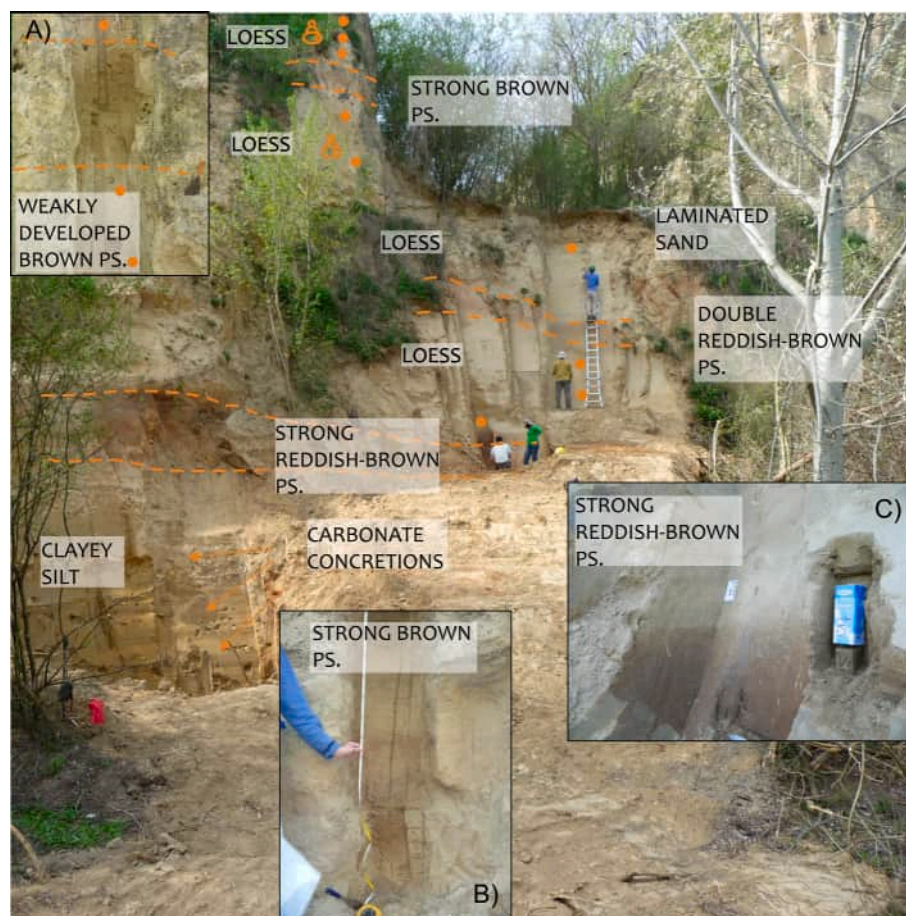


Fig. 3. The Zmajevac loess section with indicated individual horizons, stratigraphic marker horizons and sampling locations. A) The youngest, weakly developed brown palaeosol. B) strong brown palaeosol from the upper part of the section. C) The strong reddish-brown palaeosol from the oldest part of the section. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

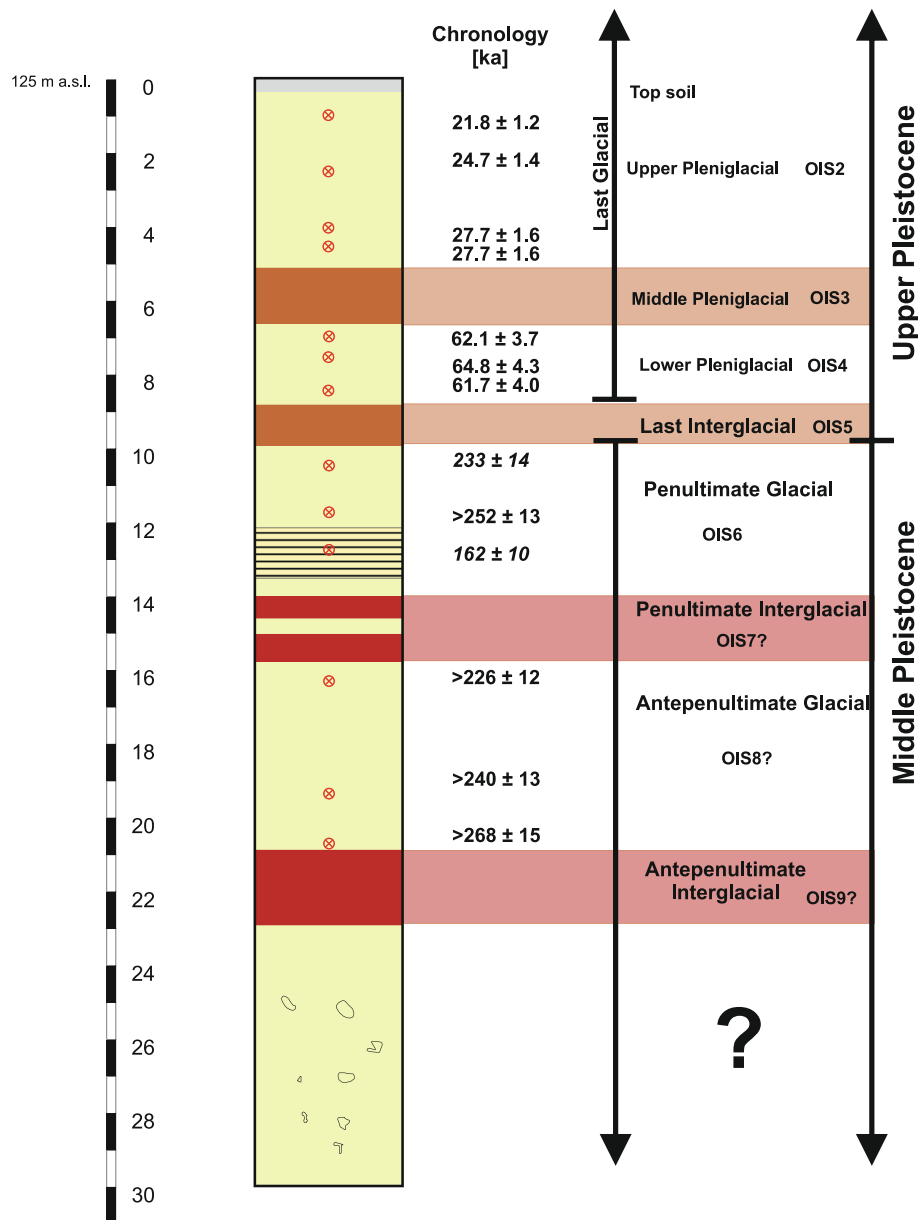


Fig. 4. The pIRIR₂₉₀ chronology of the Zmajevac loess section. Fading uncorrected ages are given for the Last Glacial loess, minimum ages are given for Penultimate Glacial (and older). The age of the laminated sand horizon was fading corrected (*in italic*).

research. The outcrop was cleaned to remove surface disturbances before any descriptions and sampling. At least 50 cm of the surface was removed from the upper part of the section, while in the lower part heavy-duty cleaning was applied (Fig. 3). Several smaller, nearby subsections were prepared for easier sampling. The profile was described, the thickness of individual horizons was measured and sampled in a 2 cm resolution for grain-size (1026 samples) and 20 cm resolution for χ and DRS analyses (115 samples). For luminescence dating 13 samples were collected using PVC and metal light-proof tubes accompanied by an additional sample from the surrounding material for gamma spectrometry. Samples were collected at approximately 80 cm intervals and bracketing the exposed palaeosols. Three samples were collected from the oldest loess, between the lowest pedocomplexes. From the same locations, separate samples for XRD analysis were collected, too. Additional samples were collected from the loess-like silty material below the lowest loess horizon for mineralogical and grain-size analyses. The sampling locations are indicated on Figs. 3 and 4, whereas the depths of the samples for luminescence dating are given in Table 1.

3.2. Luminescence dating and experimental details

Laboratory preparation and luminescence measurements were carried out under subdued red light at the luminescence dating laboratory of LIAG (Hannover, Germany). The outer 2 cm of the sample in the tube was removed to avoid possible exposure to light during and after sampling, while the rest was used for further procedure. All samples were treated with 10% HCl to remove the carbonates, Na-oxalate against coagulation and H₂O₂ to remove organic matter. The material was size separated into the fine silt fraction (4–11 μ m) using several cycles of centrifuging (Frechen et al., 1996). The fine-grained material was mounted on aluminium discs using water and then evaporated. Measurements were performed using automated Risø TL/OSL-DA-15 and -DA20 readers (Bøtter-Jensen et al., 2010) equipped with ⁹⁰Sr/⁹⁰Y β -sources, delivering 0.084 and 0.11 Gy s⁻¹. Measurements were carried out using infrared (870 $\pm$ 40 nm) LEDs. The emitted signal was filtered using the blue filter pack (Schott BG-39 and Corning 7–59).

Table 1

Results of gamma spectrometry and dose rate calculations. The error of the cosmic dose rate is 10% of the dose. Water content of $15 \pm 5\%$ was used for calculations, except for samples 3312 to 3315 where $20 \pm 10\%$ was used (after Pécsi, 1990). A systematic error of 5% was included.

Sample	Sample Id.	Depth below surface [m]	K (%)	Th (ppm)	U (ppm)	Cosmic dose rate (mGy a-1)	Total dose rate (mGy a-1)
3303	ZMA-1	1	1.89 ± 0.01	10.23 ± 0.04	3.23 ± 0.02	0.18 ± 0.02	3.99 ± 0.20
3304	ZMA-2	2.1	1.46 ± 0.01	10.09 ± 0.04	3.22 ± 0.02	0.15 ± 0.02	3.56 ± 0.20
3305	ZMA-3	4	1.49 ± 0.02	8.47 ± 0.05	2.72 ± 0.02	0.11 ± 0.01	3.21 ± 0.18
3306	ZMA-4	4.7	1.83 ± 0.01	11.28 ± 0.04	3.48 ± 0.02	0.10 ± 0.01	4.04 ± 0.21
3307	ZMA-5	6.8	1.45 ± 0.01	8.96 ± 0.04	2.71 ± 0.01	0.08 ± 0.01	3.18 ± 0.18
3308	ZMA-6	7.6	1.39 ± 0.01	9.31 ± 0.03	2.81 ± 0.01	0.07 ± 0.01	3.19 ± 0.18
3309	ZMA-7	8.5	1.72 ± 0.03	11.76 ± 0.09	3.84 ± 0.04	0.06 ± 0.01	4.08 ± 0.22
3310	ZMA-8	10.5	1.19 ± 0.01	8.07 ± 0.03	2.40 ± 0.01	0.05 ± 0.005	2.73 ± 0.17
3311	ZMA-9	11.8	1.48 ± 0.01	9.37 ± 0.04	2.38 ± 0.01	0.04 ± 0.004	3.10 ± 0.17
3312	ZMA-10	13.9	1.25 ± 0.01	11.08 ± 0.04	3.16 ± 0.02	0.03 ± 0.003	3.16 ± 0.23
3313	ZMA-11	16.5	1.54 ± 0.02	10.29 ± 0.06	2.50 ± 0.02	0.03 ± 0.003	3.11 ± 0.21
3314	ZMA-12	20.4	1.48 ± 0.02	10.51 ± 0.06	2.66 ± 0.02	0.02 ± 0.002	3.13 ± 0.21
3315	ZMA-13	20.7	1.36 ± 0.01	9.76 ± 0.03	2.55 ± 0.01	0.02 ± 0.002	2.92 ± 0.21

3.3. Dose rate determination

Samples for gamma spectrometry measurements were dried, weighted (exactly 700 g) and homogenized and then filled into Marinelli-type beakers. They were sealed using parafilm and stored for at least 4 weeks to re-establish equilibrium between radon and its daughter nuclides. Measurements were performed at LIAG in Hannover, Germany. The dose rate was calculated from obtained activities of ^{238}U , ^{232}Th and ^{40}K (Aitken, 1998). The dose rate conversion factors of Guérin et al. (2011) were applied for dose rate calculations. Alpha efficiency for polymineral fine grains of 0.08 ± 0.02 (Rees-Jones, 1995) was used. The cosmic dose rate was calculated according to Prescott and Stephan (1982) and Prescott and Hutton (1994), and an associated error of 10% is assumed. The water content was assumed to be $20 \pm 10\%$ and $15 \pm 5\%$ for the lower (samples 3312–3315) and upper (samples 3303–3311) part of the section, respectively (Pécsi, 1990). The uranium, thorium and potassium concentrations, the cosmic dose and the calculated total dose rates of the samples are given in Table 1.

3.4. High-resolution grain-size analysis and end-member modelling

For the high-resolution grain-size analysis the laser diffraction method was used (Beckman-Coulter LS 13320 PID; at LIAG, Hannover). The measurement procedure was the same as presented by Wacha et al. (2018b) and Novothny et al. (2011) i.e. the samples were not chemically treated prior to the measurements, in order not to remove detrital carbonate grains and because the organic content was insignificant in the samples. Reproducibility and reliability test were performed as well as ideal weight tests prior to the measurements. After establishing the most appropriate standard operating protocol the samples were weighted and dispersed for at least 8 h in an overhead tube rotator in a 1% ammonium hydroxide solution. Each sample was measured at least three times and the average was used for the grain-size analysis and further calculations. The granulometric parameters were calculated using the Beckman Coulter particle characterisation software. The U-ratio was calculated according to Vandenberghe et al. (1985) and considers the ratio of the coarse silt and the medium to fine silt fractions ($16\text{--}44\text{ }\mu\text{m}/5.5\text{--}16\text{ }\mu\text{m}$). To decompose the multi-modal grain size distributions into inherent subpopulations, end-member (EM) analysis was performed using the R-package EMMAgeo (Dietze and Dietze, 2019, Version 0.9.7).

3.5. Rock-magnetic measurements

For the room-temperature magnetic susceptibility measurements, the samples were dried at $30\text{ }^{\circ}\text{C}$ for 48 h, homogenized, filled in non-magnetic plastic boxes, weighed, compressed and cap sealed to prevent movement of the material. Measurements of magnetic susceptibility (χ) and its frequency dependence ($\Delta\chi$) were performed on a Magnon VFSM kappa-bridge (Magnon International, Dassel, Germany).

The frequency was set to 505 Hz for low frequency (χ_{lf}) and 5050 Hz for high frequency (χ_{hf}) magnetic susceptibility, respectively, and the field was set to 400 A/m. χ_{lf} and χ_{hf} were normalized as volume susceptibility (κ) divided by density (ρ) to the specific susceptibility with units of m^3kg^{-1} (Thompson and Oldfield, 1986). The frequency dependence was calculated by subtracting χ_{hf} from χ_{lf} :

$$\Delta\chi = \chi_{\text{lf}} - \chi_{\text{hf}} \left[\frac{\text{m}^3}{\text{kg}} \right] \quad (1)$$

and

$$\chi_{\text{fd}}[\%] = \frac{\Delta\chi}{\chi_{\text{lf}}} \times 100[\%] \quad (2)$$

Since $\Delta\chi$ is a proxy to determine the relative intensity of pedogenesis by tracking super-paramagnetic particles in the magnetic grain size around 30 nm (Peters and Dekkers, 2003), this parameter was used for comparison to the fluctuations in the LR04 stack following (Laag et al., 2021a). By recognizing similar patterns both in LR04 and $\Delta\chi$ fluctuations, 12 tie points were selected to create a correlative age model for the whole Zmajevac LPS (Table 2). This age model evaluation was carried out in R (R Core Team, 2021) by using the R-package “astrochron” (Version 0.9, Meyers, 2014). By interpolation between the tie points, for each sample a distinct age was calculated.

3.6. Bulk and clay mineralogy

The clay minerals analyses of 19 loess and palaeosol samples were performed using X-ray powder diffraction (XRPD). Preparation for the analyses included carbonate dissolution using acetic acid with ammonium acetate (1 mol/m^3) buffer of pH 5 (Jackson, 1956), grinding samples using the McCrone mill and preparation of oriented mounts

Table 2

Tie points of the correlative age model derived from the frequency dependency of magnetic susceptibility ($\Delta\chi$) compared to the LR04 stack (Lisiecki and Raymo, 2005) and the ice model by Imbrie and Imbrie (1980).

TP name	depth (m)	Age (ka)
TP_01	0.71	18
TP_02	6.01	52
TP_03	9.41	125
TP_04	10.67	140
TP_05	12.07	192
TP_06	14.17	217
TP_07	14.77	223
TP_08	15.57	240
TP_09	16.97	252
TP_10	18.67	286
TP_11	19.77	317
TP_12	21.57	333

from the $<2\ \mu\text{m}$ fraction. XRPD patterns were recorded on random mounts of bulk samples and oriented mounts. Oriented mounts of the $<2\ \mu\text{m}$ fraction were recorded after the following treatments: a) air drying, b) saturation with K^+ and Mg^{2+} c) ethylene-glycol solvation, d) Mg saturation and glycerol solvation, e) dimethyl sulfoxide (DMSO) solvation, f) heating more than $\frac{1}{2}$ hour to $400\ ^\circ\text{C}$ and $550\ ^\circ\text{C}$ (Moore and Reynolds Jr., 1997; Starkey et al., 1984). A Philips vertical goniometer (type X'Pert) equipped with a Cu tube was used with the following experimental conditions: 45 kV, 40 mA, PW 3018/00 PIXcel detector, primary beam divergence $1/4^\circ$, continuous scan (step $0.02^\circ/2\theta/\text{s}$). The interpretation of XRPD and the quantitative analysis of the bulk samples were obtained using HIGH SCORE PLUS (2016) calculation and PDF-4/MINERALS 4.5 (2020) databases. The semi-quantitative analysis of the $<2\ \mu\text{m}$ fraction was performed using the procedure described by (Bischoff, 1965).

3.7. Spectrophotometric analysis

Spectrophotometric and diffuse reflectance spectrometry (DRS) measurements were performed with a Konica Minolta spectrophotometer CM 700d at the Leibniz Institute for Applied Geophysics in Hannover, Palaeomagnetic laboratory Grubenhagen. The measurement was performed directly on the filled plastic boxes for magnetic measurements using a lens-open diameter of 0.8 cm, observer angle of 10° , and norm-light source D65. The derived data consist of values from the Lab-colour sphere, providing luminance (L^* , ranging from 0 (black) to 100 (white)), a^* (ranging from negative values (green) to positive values (red)), and b^* (ranging from negative (blue) to positive (yellow)). Additionally, backscattered reflectance values were recorded in the visible light spectrum from 400 to 700 nm in 10 nm intervals. To determine relative abundances of hematite and goethite, 1st derivative values were calculated from sample-wise measured functions of DRS following the approach of Debret et al. (2011), Laag et al. (2021a) and Wu et al. (2018), selecting the first derivative values of 435 nm and 565 nm backscattered intensity values indicative for goethite (I_{435}) and hematite (I_{565}).

3.8. Constructing the correlative age model

Variations in the northern hemisphere climate drive both the intensity of pedogenesis in loess-palaeosol sequences and also the oxygen isotope ratios in the ocean, as evidenced by e.g. the LR04 benthic $\delta^{18}\text{O}$ stack (Lisiecki and Raymo, 2005). Therefore, this marine stack can be conceptually linked to $\Delta\chi$ values to provide an age model. This is especially valuable for the older part of the section, where the uncertainty of the pIRIR₂₉₀ ages apparently increase and the method reaches its dating limit. Guided by pIRIR₂₉₀ ages, selected tie points (see Table 2) were used to quantitatively link the $\Delta\chi$ values to the age model of the LR04 stack. Data between the tie points were placed on an age scale by linear interpolation.

4. Results

Original data of this study discussed in the next chapters (grain-size, end-member modelling, environmental magnetism and DRS data) are available at Mendeley Data (Wacha et al., 2020).

4.1. Post-IR IRSL measurements of feldspar

To obtain reliable equivalent doses (D_e) used for the calculation of ages, the post-IR IRSL protocol with an elevated temperature stimulation of $290\ ^\circ\text{C}$ (pIRIR₂₉₀) was used (Thiel et al., 2011). This measuring principle was selected for two reasons. Firstly, it was proven that it is reliable for dating of older material i.e. sediments older than the Last Interglacial (e.g. Li et al., 2018; Murray et al., 2014; Schmidt et al., 2014; Thiel et al., 2011, 2014) and secondly, it was shown that luminescence

signals from feldspars obtained from elevated stimulation temperatures (e.g. 225 or $290\ ^\circ\text{C}$) fade less compared to the standard IRSL signal obtained at $50\ ^\circ\text{C}$. Fading (anomalous fading – Spooner, 1994; Wintle, 1973) is an athermal loss of signal with time observed in feldspar grains, which causes age underestimation. It can be circumvented by testing the samples for fading (Auclair et al., 2003) and correcting them using procedures proposed by Huntley and Lamothe (2001) for younger samples and Lamothe et al. (2003) or Kars et al. (2008) for older samples.

The elevated temperature post-IR IRSL (or pIRIR) measurements of the polymineral fine-grained fraction were performed as presented for the first time by Thiel et al. (2011) and subsequently successfully used at many other study sites (e.g. Thiel et al., 2014; Wacha et al., 2016, 2018a). This protocol utilizes a preheat of $320\ ^\circ\text{C}$ for 60s, before an IR stimulation for 100 s on $50\ ^\circ\text{C}$ and a subsequent IR stimulation on a significantly higher temperature, i.e. on $290\ ^\circ\text{C}$ for 200 s. The first IR stimulation step is used for recombination of the near-neighbour trap/centre pairs which are prone to fading while the IR stimulation at $290\ ^\circ\text{C}$ allows the more stable distant trap/centre pairs to recombine (Thiel et al., 2014).

The measured signals were analysed using Analyst 4.31.9 (Duller, 2015). The D_e values were obtained by integrating the first 3 to 5 s (depending on the sample) of the IRSL decay curve after fitting the dose response curve to a single saturating exponential. The signal of the final 50 s (for younger samples) and 15 s (for older samples) of the decay curve were used for background subtraction. Representative dose response and decay curves for the pIRIR₂₉₀ signal of a young (3304) and an old (3315) sample are given in Fig. 5. Most tested aliquots and aliquots used for D_e measurements showed recycling ratio close to 1 ± 0.1 and recuperation values $<3\%$. Dose recovery tests were performed to all stratigraphically youngest samples (3303–3309) and to sample 3314 from the lower part of the section. The dose recovery test is useful to check whether a given dose can be recovered successfully. In case the measured to given dose ratio is close to unity ($\pm 10\%$), the measurement protocol is considered to be appropriate and the sample datable. Doses were recovered from previously bleached aliquots (four hours in a SOL2 solar simulator) to which a dose similar to the natural was given. Residuals were also measured on bleached aliquots and were subtracted from the dose recovery ratios. The dose recovery ratios are satisfactory for the younger samples while the older samples show overestimation. Poor dose recovery tests and high residual doses for the pIRIR₂₉₀ signal in samples close to saturation were reported in several studies (e.g. Schatz et al., 2012; Thiel et al., 2014; Wacha et al., 2018a).

To investigate whether our samples are prone to fading and whether this could cause age underestimation, fading tests were performed on three aliquots of four youngest (3303–3306) and three oldest samples (3312, 3313 and 3315). Fading tests were performed according to Auclair et al. (2003) which understands repeated measurements of the sensitivity corrected luminescence intensity after increasing storage times. The given doses were $\sim 100\ \text{Gy}$ for younger and $\sim 200\ \text{Gy}$ for older samples. Since the data were significantly scattered, we calculated a mean fading rate ($g_{2\text{days}}$ -values) for each fading experiment. The young samples yielded a negative fading rate ($-0.61 \pm 0.36\%/ \text{decade}$), which is consistent to zero with $2\text{-}\sigma$ uncertainty. For the older samples (3312, 3313 and 3315), the calculated fading rate is $0.83 \pm 0.33\%/ \text{decade}$. For the six older samples (3310–3315) the natural signal saturation ratio was calculated for both without fading correction, $(n/N)_{\text{nat}}$, and after the fading correction, $(n/N)_{\text{corrected}}$ following Lamothe et al. (2003) (Table 3). After the fading correction, all samples except 3310 and 3312 are also close to saturation, showing $(n/N)_{\text{corrected}} > 0.86$, which means that the D_e values of these samples are above $2D_0$ after the fading correction and indistinguishable from saturation. For these samples, the minimum age calculated from $2D_0$ divided by the dose rate, was given as the final age estimate (Table 3). Only the sample 3310 and 3312 gave a finite fading corrected age of $233 \pm 14\ \text{ka}$ and $162 \pm 10\ \text{ka}$, respectively (Table 3, values in italic).

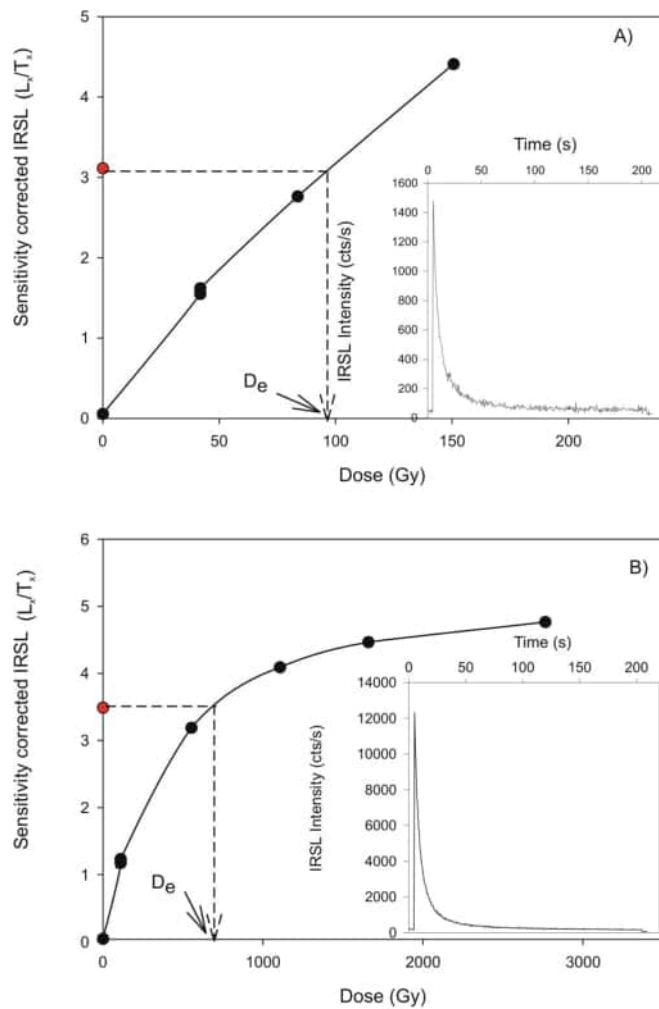


Fig. 5. Decay and Growth curves of an aliquot of a young (3304) and an old (3315) sample.

The chronological framework of the Zmajevac loess section discussed within this paper is based mainly on the fading uncorrected pIRIR₂₉₀ ages, especially for the younger part of the section. The samples older than the last interglacial, which were close to saturation, gave dose recovery tests overestimation and a small fading rate, which implies to the possibility of even older ages. For these samples, the minimum age was considered. For samples 3310 and 3312, which did not show saturation of the signal, the age was corrected for fading. The obtained ages and the stratigraphy of the Zmajevac section is given in Fig. 4.

4.2. Grain size properties

A selection of most representative differential grain-size curves of the measured samples are given in Fig. 6. The shapes of the presented frequency distribution curves of selected loess samples are in accordance with typical loess grain-size curves from the region, e.g. from Hungary (Novothy et al., 2011; Újvári et al., 2016a) or Croatia (Wacha et al., 2018a). The curve of the loess-like material at the bottom of the section is identical to the curves of selected loess samples (Fig. 6a) and are mostly unimodal with a tail in the fine-grain fraction. Some samples show a small peak in the fine sand part of the curve (Fig. 6a). Such grain-size distribution is characteristic for loess (Pécsi, 1993; Pye, 1987; Vandenberghe, 2013 – sediment type 1.b., subgroup 1.b.2). Compared to the loess curves, the palaeosol curves are polymodal with a larger proportion in the fine-grained part (<10 µm) which is due to the

Table 3
Summary of D_e values, measuring properties, dose rates and calculated pIRIR₂₉₀ ages. The fading rates (g-values) of measured samples are given and sample 3312 was corrected for fading (in *italic*). Saturation ratio (n/N) of older samples was checked and a minimum age based on $2D_0$ values was calculated for samples 3311, 3313–3315.

Sample	Sample ID	D_e (Gy)	Aliquots	Dose recovery	Residuals (ka)	Dose rate (mGy/ka)	Fading uncorrected age (ka)	(n/N) _{sat}	g_{2days} (%/decade)	(n/N) _{corr}	$2D_0$ (Gy)	Age min (ka)
ZMA-1	3303	86.85	8/8	0.92	0.05	3.98 ± 0.20	21.80 ± 1.2	–	–	–	–	–
ZMA-2	3304	87.88	8/8	0.97	0.14	3.56 ± 0.20	24.70 ± 1.4	–	–0.62	–	–	–
ZMA-3	3305	89.09	8/8	0.98	0.05	3.21 ± 0.18	27.74 ± 1.6	–	–0.62	–	–	–
ZMA-4	3306	111.76	8/8	0.93	0.03	4.04 ± 0.21	27.65 ± 1.6	–	–0.62	–	–	–
ZMA-5	3307	197.83	8/8	0.97	0.07	3.18 ± 0.18	62.13 ± 3.7	–	–	–	–	–
ZMA-6	3308	206.95	12/12	0.98	0.02	3.19 ± 0.18	64.81 ± 4.3	–	–	–	–	–
ZMA-7	3309	251.69	8/3	1.32	0.11	4.08 ± 0.22	61.73 ± 4.0	–	–	–	–	–
ZMA-8	3310	524.11	36/26	0.99	–	2.73 ± 0.17	194 ± 17	0.79	–	0.85	663 ± 6	233 ± 14
ZMA-9	3311	722.10	23/20	0.99	–	3.10 ± 0.17	210 ± 14	0.81	–	0.87	782 ± 7	>252 ± 13
ZMA-10	3312	487.94	38/30	0.99	–	3.16 ± 0.23	139 ± 10	0.69	0.83	0.74	793 ± 11	162 ± 10
ZMA-11	3313	868.90	32/73	0.99	–	3.11 ± 0.21	252 ± 20	0.89	0.83	0.95	740 ± 5	>226 ± 12
ZMA-12	3314	822.03	38/63	1.71	0.08	3.13 ± 0.21	227 ± 17	0.85	–	0.91	790 ± 11	>240 ± 13
ZMA-13	3315	797.79	40/32	–	–	2.92 ± 0.21	225 ± 16	0.81	0.83	0.87	823 ± 14	>268 ± 15

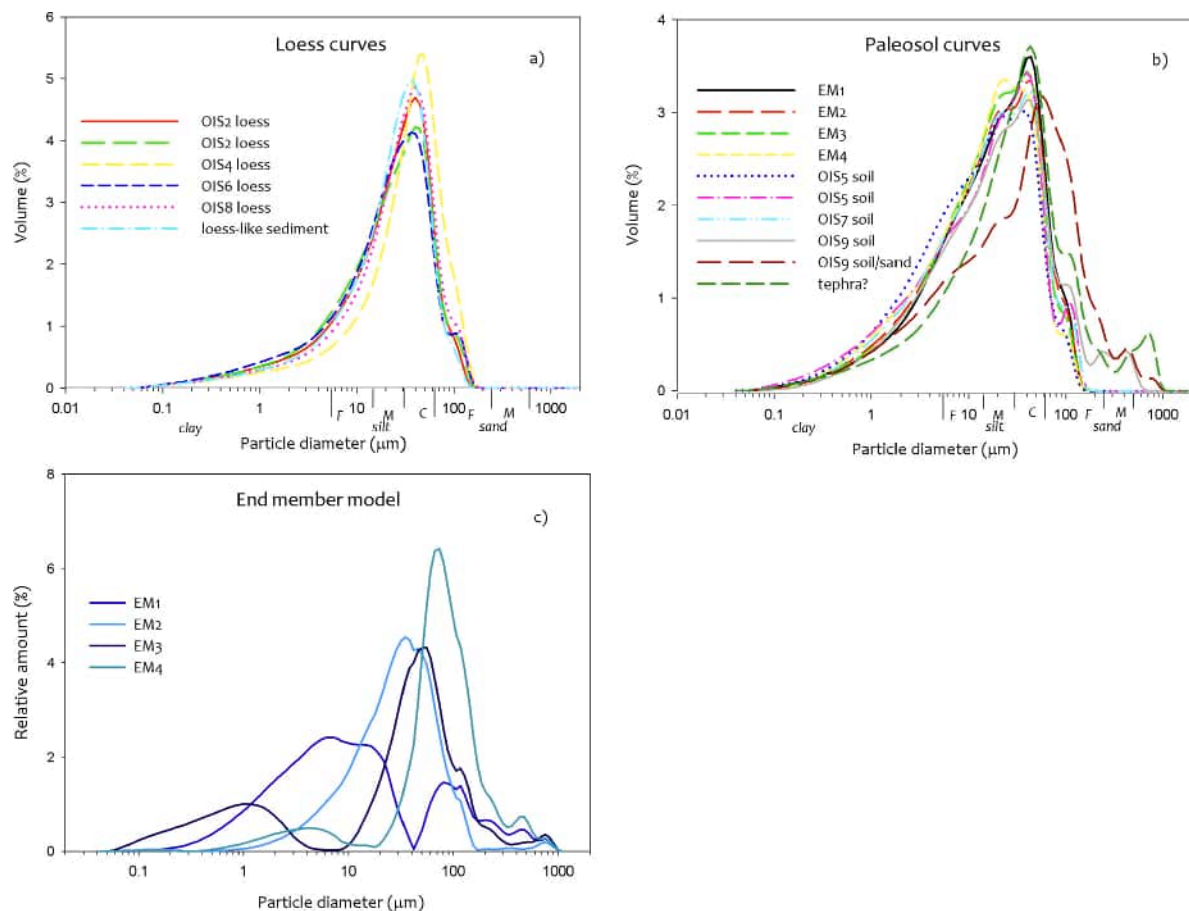


Fig. 6. The grain size curves of selected loess and palaeosols samples. a) Loess grain-size curves from loess horizons representing glacial periods (OIS2, 4, 6 and 8). The curve of the sample collected from the loess-like sediment at the base of the Zmajevac section is presented for comparison; b) Grain-size curves of different palaeosols representing interglacial (OIS 5, 7 and 9) and interstadial periods (OIS3). The curve of the distinguishable peak in grain-size on Fig. 4 is presented, as well. The peak could represent a tephra layer. c) Four endmembers obtained using the EMMAgeo package for end member modelling (Dietze and Dietze, 2019). The brighter the colour, the coarser are the grains.

increased amount of clay size particles resulting from soil-forming processes. The palaeosol curves show the same dominant mode (40 μm) as the loess but also show a second, less prominent peak at ~20 μm (Fig. 6b). A fine-grained population in loess sediments with the modal grain-size of 25–31 μm (1.b.3 subgroup) was recognized by Vandenberghe (2013) in European loess as well as in Red Clays in China and Europe. Several smaller peaks are recognized in the sand fraction. The oldest samples i.e. the OIS9 palaeosol shows several peaks in the sand part and a general increase in the grain size. Its mode is shifted towards the fine and medium sand content while some amount of coarser sand is present as well (Fig. 6b). According to Vandenberghe (2013), the 1.b loess sediments are the most common type of loess sediments, and were deposited during most severe cold periods, and provide an orbital signal. Furthermore, he concluded that they were transported in short-term, near-surface to low suspension clouds. Vandenberghe (2013) also assumed a general source region of all 1.b components to be fluvio-glacial and alluvial plains and alluvial fans of rivers originating from glaciated mountainous areas (e.g. the Danube and the Drava originating from the Alpine region). These rivers were the main sediment carriers during glacial-interglacial changes. The vertical distribution of the mean grain size (Ms) of the Zmajevac section is given in Fig. 7. A general fining upwards trend is recognized within the record. The Ms. of the entire section ranges from 19.7 μm to 79.3 μm, with an average value of 33.5 μm while the median grain-size (Md) ranges from 12.5 μm to 45.5 μm, with an average value of 24.4 μm. Both average values represent the medium silt fraction. The mean value curve pattern (Fig. 7) is similar to the >63 μm curve pattern which implies that silt-sized particles

predominate in the sediment. The content of particles <5.5 μm (clay-sized particles) ranges from 9.4% to 30.8%, with an average value of 17.6%. The content of the sand fraction (>160 μm) is up to 13%. Coarse sand (630–2000 μm), present in small amounts (up to 2%), is present only in the OIS 9 palaeosol. A more prominent increase of coarser grain sizes is visible within the OIS9 soil and the covering loess as well as in the OIS5–4 transition and OIS4 loess. An increase in grain-size during OIS6–4 compared to OIS2 was also recognized in the Süttő section in Hungary (Novothy et al., 2011) and in the Surduk section in Serbia (Antoine et al., 2009). Within the OIS8 loess, there is a pronounced peak in the mean grain size and coarse fraction (Fig. 7), which may relate to a tephra layer.

The U-ratio is a useful proxy for discriminating cold periods (high U-ratio), characterized by a dynamic aeolian environment (strong winds), from warm periods (low U-ratio), with weak winds (Vandenberghe et al., 1985). The U-ratio obtained from the Zmajevac samples range from 1.5 to 5.2 which is relatively high compared to some loess sections in the region (Obrecht et al., 2014, 2016; Vandenberghe et al., 2014). The U-ratio is highest for the OIS4 loess as well as for OIS8 loess (Fig. 7) which could imply to the accumulation of coarser grain sizes probably due to increased aridity and stronger wind activity. But it is more likely that the proximity of the Danube as a sediment source had an important influence in the contribution of coarser grain sizes (e.g. Zeeden et al., 2016). This is also supported by the laminated horizon intercalated within the penultimate glacial loess. The clay content (<5.5 μm) spans from 9.4 to 30.8% with highest rates related to the palaeosol layers and is a result of weathering and soil-forming processes (e.g. Taheri et al., 2020).

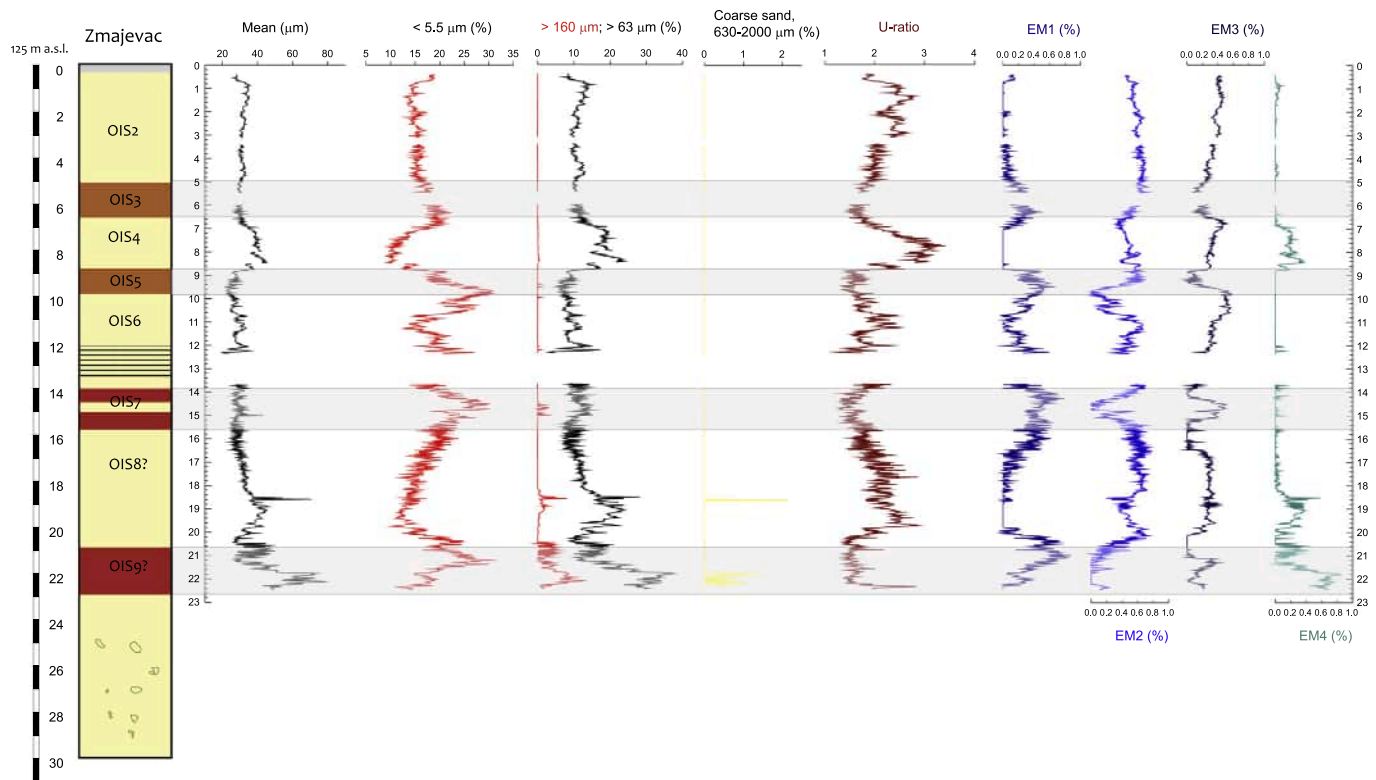


Fig. 7. Variations with depth of the grain-size content and some grain-size parameters. The samples were taken in 2 cm resolution. The loess-palaeosol section of Zmajevac is given on the left with the interpreted stratigraphy based on pIRIR₂₉₀ ages presented in Fig. 5. Loadings of the four endmembers obtained from end member modelling (Dietze and Dietze, 2019) are given for comparison.

4.3. End-member modelling analysis

EM modelling uses 4 EMs here, reflecting variations in grain sizes for distinct grain size distributions. The modelled end-members are shown on Fig. 6c which displays the contribution of all grain size classes to the modelled EMs. EM1 has two major contribution groups of two grain size classes of the fine and medium silt particles. EM1 shows another, less pronounced peak in the very fine sand range. EM2 and EM3 reflect grain size classes ranging in the medium and coarse silt fraction, however, EM3 has an additional component of fine particles (clay size particles). EM4 comprises the highest contribution of coarser grain sizes including very fine and fine sand. The contributions of EMs through time, derived from the integrated age model are plotted on Fig. 7 for comparison with grain size proxies.

4.4. Bulk and clay mineralogy

Table 4 shows the quantitative mineral composition of the analysed bulk samples. The main mineral components of the analysed loess and palaeosol samples are quartz and clay minerals while plagioclase is present in a lesser quantity. Calcite and/or dolomite are present in a smaller quantity in all samples except in the sample collected from the double reddish-brown palaeosol (sample ZmF 220–230). In some samples, a negligible number of amphiboles and K-feldspar are also present. Table 4 also shows the semi-quantitative composition of the fraction of less than 2 μm. Illitic material, chlorite and smectite are present in all samples while kaolinite and quartz are present in all samples in a lesser quantity. These findings are in general agreement with observations made by Újvári et al. (2012) with samples stemming from the Carpathian Basin. Vermiculite is uncertainly determined in only a few samples (Table 4). An example of parallel XRD patterns of an oriented and treated sample of the strong reddish-brown palaeosol is given in Supplement 2 (sample ZmG 200–210).

4.5. Rock magnetic properties

As is common for Eurasian loess deposits, χ_{lf} and $\Delta\chi$ are increased in palaeosols and low in pure loess units. The highest χ_{lf} and $\Delta\chi$ values are present in OIS 9, while OIS 7, OIS 5, and OIS 3 are less expressed. The lowest susceptibility values in a soil formation phase are reached in the palaeosol in OIS 3. The palaeosol of OIS 7 is bipartite and shows a general pattern known from other loess-palaeosol sections in the Danube region (e.g. Marković et al., 2009; Újvári et al., 2014b; Zeeden et al., 2016). The younger sub-palaeosol reflects higher susceptibilities than the older part. Magnetic enhancement is well expressed in palaeosol samples. However, pure loess units show no or little enhancement. The background-susceptibility (BGS) of $18.0 \times 10^{-8} \text{ [m}^3\text{kg}^{-1}\text{]}$ (Fig. 8) obtained from the Zmajevac section is in excellent agreement with $18.2 \times 10^{-8} \text{ [m}^3\text{kg}^{-1}\text{]}$ from Dunaszekcső, Hungary (Újvári et al., 2016a), with $16.1 \times 10^{-8} \text{ [m}^3\text{kg}^{-1}\text{]}$ for the Sémelac loess section from south-east Romania (Zeeden et al., 2016; Fig. 8) and in good agreement with BGSs calculated from north of Prague ($22 \times 10^{-8} \text{ [m}^3\text{kg}^{-1}\text{]}$) and Zemechy ($23 \times 10^{-8} \text{ [m}^3\text{kg}^{-1}\text{]}$) (Forster et al., 1996) as well as Zemun ($24.6 \times 10^{-8} \text{ [m}^3\text{kg}^{-1}\text{]}$) (Laag et al., 2021a). These values are in accordance with the BGS values of $17 \pm 2 \times 10^{-8} \text{ m}^3\text{kg}^{-1}$ proposed by (Forster et al., 1994) for central Asian and Chinese Loess-Palaeosol sequences (Fig. 8). In conclusion, the magnetic susceptibility can be regarded as an excellent recorder of the intensity of pedogenesis here.

4.6. Colourimetric analysis

The parameters luminance (L^*) and redness (a^*) from the Lab color-sphere fluctuate synchronously with changes in the stratigraphy of the Zmajevac section. L^* values are low in palaeosols, and may be reduced by mineralized organic matter, while a^* values are high in palaeosols, indicating intense weathering. Both colorimetric parameters fluctuate with the beat of the alternation of interglacials and glacials. For calculations of

5. Discussion

5.1. Chronology and stratigraphy of Zmajevac

A previous geochronological study of the Zmajevac loess section (Galović et al., 2009) was based on IRSL data. The results implied that at least two glacial cycles are archived at Zmajevac. The oldest obtained age of the lowermost dated sample at Zmajevac was 217 ± 22 ka (Galović et al., 2009). Nevertheless, the obtained ages were ambiguous and the correlation with other nearby dated sections was not straightforward. It should be also considered that within the study of Galović et al. (2009) the multiple aliquot additive dose (MAAD) protocol was used for obtaining the equivalent doses (D_e) as well as no fading corrections were performed. Hence the obtained data must be underestimated. Nevertheless, the data presented within this study partly support the first reconstructions by Galović et al. (2009) and give a more detailed and robust chronological framework.

The presented pIRIR₂₉₀ ages clearly show that at Zmajevac, a sedimentary archive of at least three glacial-interglacial cycles is exposed. The Zmajevac loess section starts with a horizon of clayey silt rich in carbonate concretions on top of which a strong rubified brown palaeosol is developed. The age of this loess-like horizon is not known; it can be correlated to the Middle Pleistocene (> 280 ka; OIS9 or older). The lowermost palaeosol was covered with loess which gave uncorrected pIRIR₂₉₀ ages of 225 ± 16 ka (sample 3315), 227 ± 17 ka (sample 3314) and 252 ± 20 ka (sample 3313) (Fig. 4). However, after the fading correction, the natural pIRIR₂₉₀ signal of the samples are indistinguishable from saturation and only the minimum age can be given, $>268 \pm 15$ ka (sample 3315), $>240 \pm 13$ ka (sample 3314) and $>226 \pm 12$ ka (sample 3313). Nevertheless, when all data are considered we tentatively propose this soil to correlate to OIS 9 and the oldest loess to OIS 8.

Above the oldest loess horizon, the double reddish-brown palaeosol is exposed. Based on the obtained pIRIR₂₉₀ ages, the age of this soil complex is ambiguous. Below the double palaeosol (sample 3313) an uncorrected age of 252 ± 20 ka (corrected minimum age of $>226 \pm 12$ ka) was obtained from the loess. Above this soil complex laminated sand is exposed which gave a fading corrected pIRIR₂₉₀ age of 162 ± 10 ka (sample 3312). This age seems underestimated when compared with surrounding data. This age can also be correlated to OIS6, implying that the surrounding ages obtained from loess are overestimated. However, it was shown in many studies that pIRIR₂₉₀ dating protocol applied on loess samples gave reliable ages for older samples ($>OIS6$) useful for stratigraphic interpretations and correlation deposits (e.g. Thiel et al., 2014). The loess above the sand gave uncorrected ages of 210 ± 14 ka (sample 3311, fading corrected age) and 233 ± 14 ka (sample 3310, below the brown palaeosol, corrected minimum age of $>243 \pm 14$ ka). It is not quite clear what is the reason for this age inversion. A similar discrepancy of the age of the laminated horizon was observed by Fenn et al. (2020) for the Erdut section. Fenn et al. (2020) recorded an age overestimation of a sample collected from the top of the laminated sediment, which they explain due to its fluvial origin of the sand causing partial bleaching. As described in Fenn et al. (2020), it is very likely that the sand was reworked, insufficiently bleached during deposition (fluvial transport) and recently bioturbated and thus the obtained age underestimated. Nevertheless, the age of the laminated sediment from Zmajevac of 162 ± 10 ka (fading corrected age; sample 3313) correlates well with results from nearby Erdut section and implies the same conclusion as Fenn et al. (2020), a temporary shift to fluvial sedimentation. Similar was also recorded at Stari Slankamen (Marković et al., 2011).

A double palaeosol is exposed at the Šarengrad II section in Croatia. Based on pIRIR₂₂₅ ages the double palaeosol at Šarengrad II was tentatively correlated to the OIS7 (Wacha et al., 2013). Similar double soil complex is also reported in the Danube loess region; the V-S2 in the Serbian loess stratigraphy (Marković et al., 2013, 2015) and the BD₁-BD₂ in the Hungarian loess stratigraphy (Thiel et al., 2014; Újvári et al., 2016a) and is correlated to OIS7. Based on our results, for which we

acknowledge that they are ambiguous in this part of the section, we also tentatively correlate the double palaeosol to OIS7 (Fig. 4). The fading corrected age of 233 ± 14 ka (sample 3310) and fading uncorrected age of 210 ± 14 ka (sample 3311) obtained from the loess covering the laminated sand can be correlated to at least OIS6, i.e. to the Penultimate Glacial or any older glacial. But when the minimum age of $>252 \pm 13$ ka (sample 3311) is considered a correlation to OIS8 is more appropriate. Nevertheless, the data clearly shows evidence of erosional processes within this loess horizon (laminated sand horizon and reduced thickness of the covering palaeosol-OIS5). A more detailed record of OIS6 loess is exposed at Erdut (Fenn et al., 2020) giving pIRIR₂₂₅ ages ranging from 137 ± 6 ka to 158 ± 8 ka. Above the Penultimate Glacial loess (OIS6) at Zmajevac, a strong brown palaeosol is exposed bracketed with pIRIR₂₉₀ ages of 233 ± 14 ka (fading corrected age, sample 3310) and 61.7 ± 4.0 ka (sample 3309; Fig. 4). These results place the brown palaeosol into OIS5. A single thick brown palaeosol correlating to OIS5 is also exposed at the Gorjanović loess section in Vukovar (Wacha and Frechen, 2011) but the soil there has a significant thickness compared to the Zmajevac OIS5 soil. A single soil pedocomplex (S1 palaeosol after Marković et al., 2009) is correlated with the complete OIS5 period in the Danube loess stratigraphy (Marković et al., 2015). Furthermore, the results from Zmajevac are very similar when compared to the fading corrected pIRIR₂₂₅ ages from the Gorjanović loess section in Vukovar (Wacha and Frechen, 2011). In Vukovar ages around 64 ka were obtained above the OIS5 palaeosol, same as in Zmajevac (Fig. 4) but significantly younger ages were obtained for the Penultimate Glacial (OIS6) loess (141 ± 11 ka; Wacha and Frechen, 2011). The somewhat older ages of the Penultimate Glacial loess from Zmajevac directly below the OIS5 soil could be due to age overestimation but could also imply erosion, possibly due to fluvial activity and/or erosion due to its ridge position. A larger discontinuity was also recognized at the Paks loess section in Hungary (Thiel et al., 2014) where the OIS4 loess and the OIS5 soil are completely missing. However, this is not the case with the Paks cores (Novothy et al., 2020) where the last interglacial palaeosol is preserved. The difference between the Paks outcrop and core record is very likely due to their position, the Paks section being in a slope position prone to erosion. The Last Glacial loess from Zmajevac gave ages ranging from 64.8 ± 4.3 ka to 21.8 ± 1.2 ka (Fig. 4). Within the Last Glacial loess, a weakly developed brown palaeosol is intercalated. It is bracketed by pIRIR₂₉₀ ages of 62.1 ± 3.7 ka obtained below and 27.6 ± 1.6 ka above the palaeosol. These ages imply that soil-forming processes happened during the OIS3. In the wider area, OIS3 is characterized by soil formation and weakly developed soils (LIS1 in Serbian loess stratigraphy; Marković et al., 2015) or humic and weakly developed soil horizons (e.g. Novothy et al., 2011; Újvári et al., 2014b) are often recognized within the Last Glacial loess. The ages of the youngest exposed loess range from 21.8 ± 1.2 ka to 27.6 ± 1.6 ka. The obtained results also revealed that deposits younger than 20 ka are missing, which was also observed in the nearby section of Dunaszekcső (Újvári et al., 2014a, 2016b, 2017). The age of the uppermost sample is 21.8 ± 1.2 ka above which the recent soil is exposed. It is very likely that loess younger than 20 ka was pedogenetically altered to form the modern soil (Rubinić et al., 2018) but erosion and agricultural activity could be the reason for the hiatus, too. The age-depth plot resulting from the correlative age model and the pIRIR₂₉₀ ages is given in Fig. 9 while the comparison of the independently developed age models is given in Fig. 10. Fig. 9 shows the underestimation of the (Galović et al., 2009) data, especially in the older part and the dating limit of the older samples. Nevertheless, within error limits, the ages can be used as a chronological framework and for stratigraphic interpretations. To reliably establish a correlative age model, several assumptions are taken into considerations, including guidance by luminescence data, experience in the proxy patterns of soils and importantly field experience. The age model can be assumed to be correct on glacial/interglacial and possibly obliquity timescales. However, we can also expect imperfections due to (a) uncertainty of the LR04 stack on the order of 4 ka (Lisiecki and Raymo, 2005), (b) uncertainty in

the tie points due to the low-resolution magnetic susceptibility data (on the order of several ka up to ~20 ka), and (c) probably not uniform sedimentation rates between tie points. In addition, the tie point at the end of MIS 7 may be shifted towards younger ages, but the rather low-resolution χ data do not allow an unambiguous assignment.

5.2. Palaeoenvironmental evolution derived from the grain size record and the EMM

At the base of the Zmajevac loess-palaeosol section loess-like sediments are exposed on top of which the oldest (OIS9) palaeosol is formed. This palaeosol has an increased sand content which implies, same as the loess-like sediment below, that approximately during the antepenultimate glacial (OIS8) the environment in the area transitioned from an alluvial plain environment of the rivers Drava and Danube into a plateau-like, stable, aeolian depositional environment. The high contribution of EM4 indicates abundant local material. The evidence of the aeolian influence is the relatively constant influx of silt in the grain-size record (average M_s 33.5 μm ; the dominance of the silt fraction and the consistent presence of EM2 and EM3). The increased sand content in the lower part of the section, besides the vicinity of the alluvial source, may also be related to stronger winds during the OIS8. Regarding the pedogenesis, the clay content and EM1 is here considered a direct result of the pedogenesis. Our data clearly show the differentiation between cold glacial periods and the overprint of warmer climate on loess, evidenced as palaeosols. The OIS3 soil, however, shows only weak pedogenic overprinting, compared to older soils. This is a common feature in Eurasian loess with different soil formation intensity (Bronger, 1975), which is consistently weaker than interglacial soil formation. The U-ratio shows different values for loess and for palaeosols, which also indicates a difference in wind strengths over the last ~330 ka. The U-ratio values of OIS8 reflect first a rapid increase of wind strength towards younger samples until ~20 m, then a gentle decrease, indicating a progressively weakening wind strength. Wind intensities from OIS7 and 6 remain on a quasi-comparable low level. Thus, the variations of the U-ratio values remain lower in the palaeosols compared to the loess units,

indicating more constant wind intensities. Lower U-ratios in palaeosols could also be the result of post-depositional alterations due to pedogenesis. Inside the loess units from OIS 8, 6 and 4, high varying U-ratios may indicate strong wind events, which appear in a somehow cyclical shape. The overall trend of the U-ratio values from OIS6 reflect a slightly decreasing trend as well as in OIS8, but much weaker expressed. OIS4 is indicating first a rapid increase of wind intensity (between 8.5 and 9 m), then a kind of plateau where wind intensities remain strong between 8.5 and 7.5 m, followed by a well-expressed drop in wind intensities (between 7.5 and 6.5 m). However, the availability of coarser grain sizes in the lower part of the section also influences the U-ratio and interpreting the U-ratio as a result of wind strength only would be oversimplification. OIS4 U-ratio values are comparably high, similar to U-ratio values from Süttő (Novothy et al., 2011) but do not reflect any periodicity like in Süttő. The OIS2 loess seems quite uniform if the mean grain-size values are considered (Fig. 7). However, when the total sand content and the U-ratio are taken into consideration one can recognize a slight coarsening upwards trend. This is opposite of what is seen for OIS 8 and OIS4. Similar was also recognized within OIS2 loess from Süttő (Novothy et al., 2011) and Surduk (Antoine et al., 2009). OIS2 loess reflects, opposite to all other loess units, an increasing trend of wind intensities from older to younger samples. The Last Glacial Maximum (LGM - OIS2) is a period with highest mass accumulation rates in Europe (Frechen et al., 2003), due to the availability of the source material as well as the cold, dry and arid climate. Following these observations, the Zmajevac LPS provides insights into past wind intensity changes with more unstable conditions during glacials and more stable wind intensities during interglacials.

We interpret EM1 as representing mainly fine clay particles formed during pedogenesis which is also supported by a good correlation with χ_{fd} (%) (see eq. 2) to EM1 scores. The coarse component may be due to aggregation, or due to fine sand input during these intervals. The intermediate EM2 and EM3 may be interpreted as grains provided by longer-distance transport and exclusively aeolian transportation. They could also imply to local supply. EM2 is finer than EM3 and could therefore imply to even more distant sources compared to EM3. EM3 is

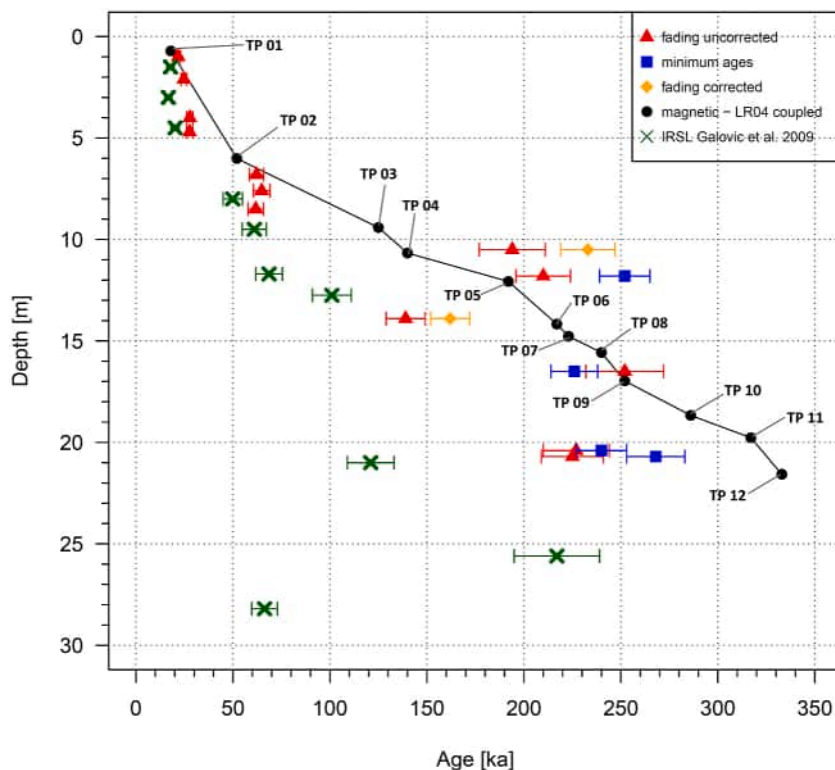


Fig. 9. Comparison of the independently developed age models for the Zmajevac LPS. Red triangles - fading-uncorrected pIRIR₂₉₀ ages, blue squares - the minimum pIRIR₂₉₀ ages; green crosses - IRSL ages from Galović et al. (2009); black filled circles - independently correlated tiepoints from Table 1. The lines in between reflect the interpolation for all samples in between the tie points. The tie points are given in Table 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

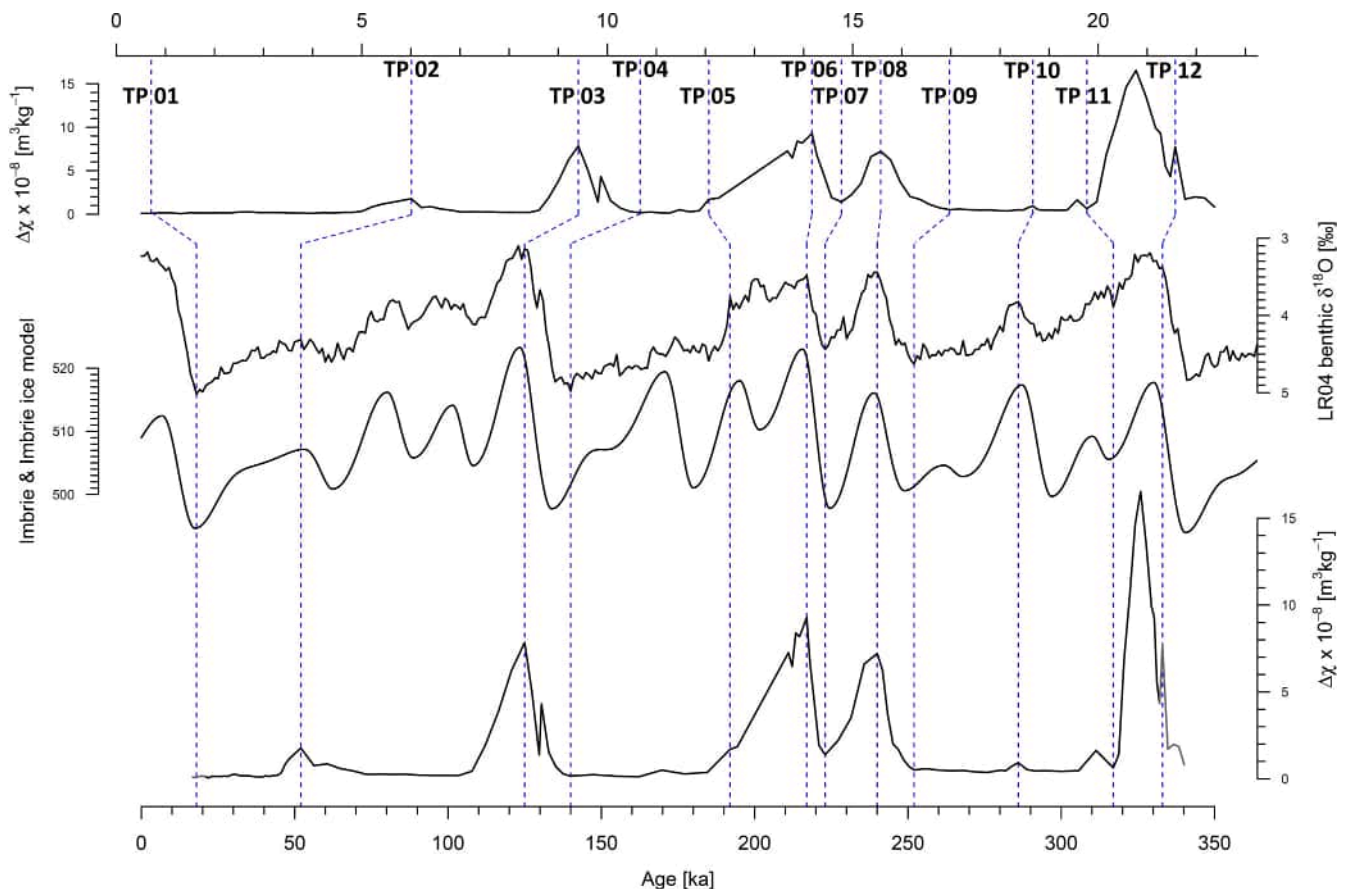


Fig. 10. The correlative age model based on the correlation between frequency dependent magnetic susceptibility $\Delta\chi$ and the LR04 stack and Imbrie and Imbrie (1980) ice model. The tie points are indicated as dashed blue lines and are compiled and labelled in Table 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

interpreted as related to silt sources in the wider area (from the direct vicinity to several km) because of its clay contribution and also fine sand component. The negative correlation of EM2 with L^* reflect low EM scores in palaeosols. Generally, dark sediment colours (low L^*) indicate the presence of palaeosol formation and can be correlated with EM2. The applied age model reflects the dependence of increased magnetic susceptibility (χ_{fd}) and its frequency dependence ($\chi_{fd} [\%]$) where large amounts of fine particles are detectable in consistent stratigraphic units. EM3 also correlates well with the clay content and the U-ratio values (Fig. 7). EM4 reflects coarse grains accumulated predominantly during glacial conditions. Especially OIS 4, OIS6 and OIS8 reflect high amounts of these coarse grain sizes. At the bottom of the Zmajevac LPS, the relative amount of coarse grains is highest. During glacial conditions, intensified wind strength is assumed triggering the increased input of coarse grains to the Zmajevac LPS. Additionally, potential sources (close to the geographical location of the Zmajevac LPS) may be available for uptake, probably alluvial deposits related to the Danube and Drava rivers in the vicinity, because a stabilizing vegetational cover is missing due to too cold and dry environmental conditions during the glacials.

5.3. Clay minerals and hematite and goethite content as indicators of provenance and climate

Mineralogical data suggest that the dominant association for loess-palaeosol sequences in Croatia, in particularly Zmajevac is garnet, minerals from the epidote-zoisite group, chlorite and amphiboles (Banak et al., 2013; Galović, 2014). Similar is in the wider Pannonian basin (Thamó-Bozsó et al., 2014). The main mineral phases in the clay mineral fraction are illitic material, smectite and chlorite (Table 4). An almost

identical mineral composition of the clay mineral fraction of two loess samples from the area of Zmajevac was also obtained by Újvári et al. (2012). The vertical distribution along the Zmajevac section is fairly uniform and their mineralogical composition is typical for loess-palaeosol sequences in the area (Markovic et al., 2004; Újvári et al., 2012; Varga et al., 2011). The presence of the illitic material and chlorite, which are well represented throughout the profile, could imply to an environment which was cold(er) with reduced chemical weathering and hence limited soil formation (Fanning et al., 1989; Wilson, 1999). A somewhat lesser amount of chlorite was detected in the Last Interglacial brown palaeosol, which probably together with a larger amount of smectite and the absence of feldspar implies to a warmer and more humid environment during the Last Interglacial. Kaolinite is uniformly, but only in subordinate amounts, distributed along the analysed profile with no obvious differences in quantity between loess and palaeosol (Table 4) which could suggest that kaolinite is mainly detrital. This is consistent with the fact that samples mainly contain kaolinite forming intercalation compounds with DMSO or well-crystallized kaolinites. Namely, according to Durn et al. (2014 and references therein) kaolinite that does not intercalate with DMSO is predominantly of paedogenic origin, while kaolinite that forms intercalations with DMSO is inherited from the kaolinite-containing parent rock. Table 4 shows that smectite is more abundant in palaeosol samples as well as in Penultimate Glacial loess, compared to other loess samples, i.e. there is an increased smectite content in the older part of the section. The presence of increased smectite content in the Penultimate Glacial loess and -Interglacial palaeosols probably indicates multiple reworking and resedimentation of these horizons. Generally, there are two main sources of smectite in soils; smectite can be of detrital origin or it can form by alteration of

volcanic material, micas, feldspars and various Fe—Mg silicates (Millot, 1964; Taheri et al., 2020; Weaver, 1989). It can also form by neoformation, by precipitation directly from the solution. According to (Weaver, 1989), climatic and topographic conditions necessary for the formation of smectite are low relief and/or low permeability, low rainfall and low temperature. A part of the smectite in the analysed palaeosols probably has a detrital origin, but a part could have been created by the alteration of mica and/or feldspars, and/or the amphibole-pyroxene group of minerals. The reason for the very low content or the complete absence of vermiculite in the analysed samples (Table 4) could be their alteration into smectite. Dioctahedral vermiculite is more common in soils than trioctahedral vermiculite (Jackson, 1957). Due to continued weathering of minerals such as biotite, trioctahedral vermiculite can be altered in dioctahedral form, but since the charge is usually lowered it alters into smectite (Weaver, 1989). These results are in accordance with (Bronger, 2003) who reported that high amounts of vermiculite are present in loess-palaeosol sequences from Central Asia, whereas in loess-palaeosol sections from Hungary, Serbia and Croatia illite and smectite are dominant mineral phases. Markovic et al. (2004), Galović et al. (2006), Varga et al. (2011) and Újvári et al. (2012; 2014) also presented a similar clay minerals composition (illite, chlorite and smectite) determined in studied loess-palaeosol sequences from Hungary, Serbia and Croatia.

Since the initial work of Deaton and Balsam (1991), distinct back-scattered wavelengths are indicative for the relative abundance of hematite and goethite (e.g. Wu et al., 2018), both high coercive magnetic minerals. This dependence was shown by a recent study from Laag et al. (2021a), by showing high correlations between the mentioned wavelength bands with high-coercivity magnetic parameters. For hematite the favoured forming conditions are precisely investigated and characterized, providing valuable information about the past in terms of temperature, precipitation, and seasonality regimes. Hematite contains only the oxidized form of iron. Its formation is favoured by both a short period of limited precipitation and high temperatures (Maher, 1998).

The limit of precipitation is considered as the ideal formation condition, and usually is below 350 mm/year (Ji et al., 2001, 2002). In relation to magnetite and maghemite, the enhancement of hematite content is positively influenced by a longer warmer and dryer period following a shorter period of soil wetness. It forms most effectively when the period of soil wetting is too short to produce significant contents of magnetite and maghemite, which leads (regarding the mentioned conditions) to reduced magnetic susceptibility (Ji et al., 2001). Additionally, a rapid release of iron under neutral or slightly acidic conditions influences the formation of hematite positively, which forms probably from dehydration of ferrihydrite (Balsam et al., 2004). The determination of goethite content follows the same procedure as hematite with other back-scattered wavelengths. However, the climatic interpretation of the goethite content is more uncertain because formation conditions occur in wide temperature and precipitation regimes. Commonly, the goethite formation is connected to lower temperatures and/or higher (less seasonal) precipitation than favoured by the formation of hematite. Additionally, the goethite formation is favoured under higher organic matter content, lower pH and increased aluminium concentration (Balsam et al., 2004).

Diffuse reflectance spectrometry provides insights into relative abundances of hematite and goethite, whose favoured conditions for neoformation is well-known. Luminance (L^*) reflects darkness in palaeosols and bright colours in pure, unweathered loess units and correlates negatively with $\Delta\chi$ values. The high L^* values in loess represent higher amounts of bright minerals like quartz and carbonates. Darker material (reduced L^*) is commonly found in palaeosols, mostly influenced by weathered organic matter and changed mineral composition. Generally, in Fig. 11 the frequency dependent susceptibility, goethite (green), and hematite (red) are displayed on an age scale produced by the age model comparing LR04 and depth-based $\Delta\chi$ values. Reflectance values, indicating goethite abundance, show correlation with $\Delta\chi$ values ($r = -0.81$, $p(a) < 2.2e-16$, $n = 109$). Generally, goethite abundances are the lowest, where palaeosols are indicated by high $\Delta\chi$

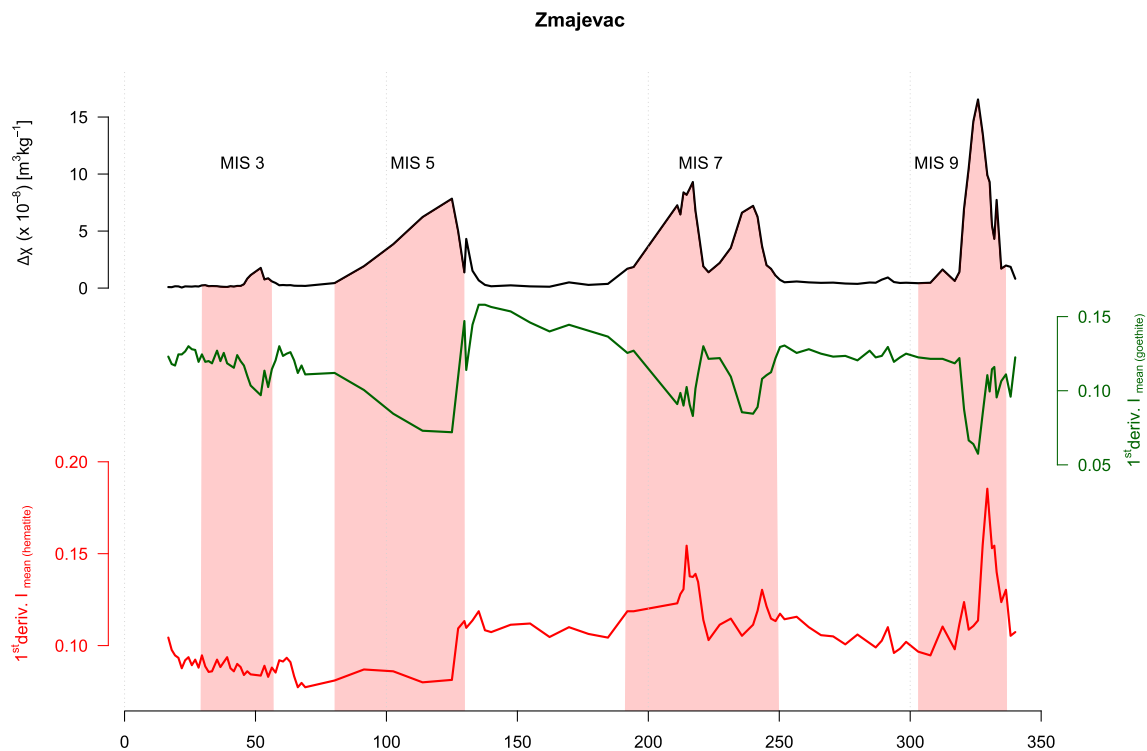


Fig. 11. Frequency dependence of magnetic susceptibility ($\Delta\chi$, black), first derivative of DRS intensity wavelength 435 nm (green, indicative for goethite) and 565 nm (red, indicative for hematite) on an age scale derived from the tie points of the correlative age model (Table 1). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

values. Hematite is increased in warm (uneven) OISs, except for OIS3 and 5, and decreases from the oldest samples to the youngest ones. The highest abundance of hematite is in OIS9 palaeosol. Goethite fluctuations of OIS 7 are divided into an older (badly expressed) peak and younger peak, which are well expressed and timed, where the $\Delta\chi$ values are the highest. A clear and permanent drop in hematite content appears at the beginning of OIS 5, where the lowest hematite contents are displayed, which is unusual and interesting regarding the palaeoclimatic implications during OIS 5. In OIS 3, the hematite content is increasing again. Commonly, OIS 9 and 5 are the most intense interglacials in the last 350 kyr (PAGES, 2016). At Zmajevac OIS 9 fits this pattern, whereas OIS 5 does not. The decreasing trend from older to younger samples of the HGR was observed in the Zemun LPS (Laag et al., 2021a) and interpreted as decreasing of summer temperatures. While at Zemun a very slight decreasing trend of susceptibility parameters is observed in palaeosols from older to younger samples, Zmajevac indicates an overall, clearly visible decreasing trend.

6. Conclusion

The Zmajevac loess-palaeosol section was re-examined in high resolution to get insight into the palaeoclimatic history of the region and to update the chronological framework. A detailed and robust chronology is given, established based on post-infra-red stimulated luminescence (IRSL) dating with luminescence signals registered at 290 °C (post-IR IRSL₂₉₀ or pIRIR₂₉₀). The obtained results show that, contrary to previous interpretations, at least three glacial-interglacial cycles are preserved at Zmajevac. While the pIRIR₂₉₀ ages of the older samples are somewhat ambiguous, the younger samples are in stratigraphic order. The correlative age model, which is based on $\Delta\chi$ to LR04 correlations, generally supports the luminescence-based stratigraphy of Zmajevac.

Changes in grain-size were analysed with respect to supposed atmospheric dynamics of the area and the palaeoclimate. Four grain size EMs were identified. We interpret EM1 as representing mainly fine silt and clay particles formed during pedogenesis which is also supported by a good correlation with χ_{fd} (%). The intermediate EM2 and EM3 may be interpreted as grains provided by longer-distance transport and exclusively aeolian entrainment. EM4 reflects coarse grains accumulated predominantly under glacial conditions. The EM-s distinguish grain-size subpopulations which may imply variations in the provenance of the material. Amplitude and variations of the U-ratio inside palaeosol- and loess units indicate differences in wind intensity and changes of individual and high-intensity wind events in interglacials and glacials as well as changes in the distance to the source area. Loess units witness increased wind intensities with intercalated high-intensity events whereas palaeosols, in contrast, reflect milder, less windy and more stable conditions. Besides the wind intensity, the increase in grain size visible in the older part of the section also indicates the shorter distance of the source area, which are the nearby alluvial deposits of the Drava and Danube.

The mineralogical composition of the Zmajevac section consists dominantly of quartz, clay minerals, plagioclase and calcite. The distribution of the clay minerals is fairly uniform, consisting dominantly of smectite, illitic material and chlorite, which is comparable with earlier studies of loess-palaeosol sequences in the SE part of Central Europe/Danube loess region. The presence of the increased smectite content in the Penultimate Glacial loess and -Interglacial palaeosols probably indicates multiple reworking and re-sedimentation in this part of the section. Smectite in palaeosols is probably in part of a detrital origin, but a part likely originated from micas and feldspars as a result of weathering.

Changes both in magnetic susceptibility and colourimetric parameters indicate a strong decrease of summer heat and seasonality over the last three interglacial/glacial cycles. Additionally, the decreasing susceptibility parameters may indicate an increased limitation of the palaeoprecipitation. The reason for this decreasing trend remains

questionable and we propose to investigate the Zmajevac section for further colourimetric and rock-magnetic parameters to find an explanation for this trend. The combination of grain size analysis with further colourimetric and rock-magnetic investigations may reveal a direct link of temperature, precipitation and wind-strength seasonality, possibly linked to small millennial-scale climatic changes.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.palaeo.2021.110504>.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by the Croatian Science Foundation (HRZZ, UIP-2013-11-4425 *Standardisation and Applied Investigation of Quaternary Sediments in Croatia*). Special thanks to technicians from S3 and S5 from LIAG, for all the laboratory work; Gudrun Drewes, Astrid Jaeckel, Petra Posimowski, Sonja Riemenschneider and Kathrin Worm. Thanks to Mihovil Brlek for help during sampling. This is IPGP contribution 4219.

References

- Aitken, M., 1998. *An Introduction to Optical Dating*. Oxford University Press, Oxford.
- Antoine, P., Rousseau, D.-D., Fuchs, M., Hatté, C., Gauthier, C., Marković, S.B., Jovanović, M., Gaudenyi, T., Moine, O., Rossignol, J., 2009. High-resolution record of the last climatic cycle in the southern Carpathian Basin (Surdok, Vojvodina, Serbia). *Quat. Int.* 198, 19–36. <https://doi.org/10.1016/j.quaint.2008.12.008>.
- Antoine, P., Coutard, S., Bahain, J., Locht, J., Hérison, D., Govaal, E., 2021. The last 750 ka in loess–palaeosol sequences from northern France: environmental background and dating of the western European Palaeolithic. *J. Quat. Sci.*, jqs.3281 <https://doi.org/10.1002/jqs.3281>.
- Auclair, M., Lamothe, M., Huot, S., 2003. Measurement of anomalous fading for feldspar IRSL using SAR. *Radiat. Meas.* 37, 487–492. [https://doi.org/10.1016/S1350-4487\(03\)00018-0](https://doi.org/10.1016/S1350-4487(03)00018-0).
- Balsam, W., Ji, J., Chen, J., 2004. Climatic interpretation of the Luochuan and Lingtai loess sections, China, based on changing iron oxide mineralogy and magnetic susceptibility. *Earth Planet. Sci. Lett.* 223, 335–348. <https://doi.org/10.1016/j.epsl.2004.04.023>.
- Banak, A., Pavelić, D., Kovačić, M., Mandić, O., 2013. Sedimentary characteristics and source of loess in Baranja (Eastern Croatia). *Aeolian Res.* 11, 129–139. <https://doi.org/10.1016/j.aeolia.2013.08.002>.
- Basarin, B., Bugge, B., Hambach, U., Marković, S.B., Dhand, K.O., Kovačević, A., Stevens, T., Guo, Z., Lukić, T., 2014. Time-scale and astronomical forcing of Serbian loess–paleosol sequences. *Glob. Planet. Chang.* 122, 89–106. <https://doi.org/10.1016/j.gloplacha.2014.08.007>.
- Biscaye, P.E., 1965. Mineralogy and Sedimentation of recent Deep-Sea Clay in the Atlantic Ocean and Adjacent Seas and Oceans. *Geol. Soc. Am. Bull.* 76, 803. [https://doi.org/10.1130/0016-7606\(1965\)76\[803:MASORD\]2.0.CO;2](https://doi.org/10.1130/0016-7606(1965)76[803:MASORD]2.0.CO;2).
- Bokhorst, M.P., Vandenberghe, J., Stümege, P., Lanczont, M., Gerasimenko, N.P., Matviishina, Z.N., Marković, S.B., Frechen, M., 2011. Atmospheric circulation patterns in central and eastern Europe during the Weichselian Pleniglacial inferred from loess grain-size records. *Quat. Int.* 234, 62–74. <https://doi.org/10.1016/j.quaint.2010.07.018>.
- Bøtter-Jensen, L., Thomsen, K.J., Jain, M., 2010. Review of optically stimulated luminescence (OSL) instrumental developments for retrospective dosimetry. *Radiat. Meas.* 45, 253–257. <https://doi.org/10.1016/j.radmeas.2009.11.030>.
- Bradák, B., Seto, Y., Nawrocki, J., 2019. Significant pedogenic and palaeoenvironmental changes during the early Middle Pleistocene in Central Europe. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 534, 109335. <https://doi.org/10.1016/j.palaeo.2019.109335>.
- Bronger, A., 1975. Paläoböden als Klimazeugen — Dargestellt an Löß-Boden-Abfolgen des Karpatenbeckens. *E & G Quat. Sci. J.* 26, 131–154. <https://doi.org/10.3285/eg.26.1.08>.
- Bronger, A., 2003. Correlation of loess–paleosol sequences in East and Central Asia with SE Central Europe: towards a continental Quaternary pedomorphology and paleoclimatic history. *Quat. Int.* 106–107, 11–31. [https://doi.org/10.1016/S1040-6182\(02\)00159-3](https://doi.org/10.1016/S1040-6182(02)00159-3).
- Chu, W., 2018. The Danube Corridor Hypothesis and the Carpathian Basin: Geological, Environmental and Archaeological Approaches to Characterizing Aurignacian Dynamics. *J. World Prehist.* 31, 117–178. <https://doi.org/10.1007/s10963-018-9115-1>.
- Core Team, R., 2021. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://r-project.org>.

- Deaton, B.C., Balsam, W.L., 1991. Visible spectroscopy; a rapid method for determining hematite and goethite concentration in geological materials. *J. Sediment. Res.* 61, 628–632. <https://doi.org/10.1306/D4267794-2B26-11D7-8648000102C1865D>.
- Debret, M., Sebég, D., Desmet, M., Balsam, W., Copard, Y., Mourier, B., Suspendigui, A.-S., Arnaud, F., Bentaleb, I., Chapron, E., Lallier-Vergès, E., Winiarski, T., 2011. Spectrocolorimetric interpretation of sedimentary dynamics: The new “Q7/4 diagram”. *Earth Sci. Rev.* 109, 1–19. <https://doi.org/10.1016/j.earscirev.2011.07.002>.
- Dietze, E., Dietze, M., 2019. Grain-size distribution unmixing using the R package EMMAgeo. *E & G Quat. Sci. J.* 68, 29–46. <https://doi.org/10.5194/egqsj-68-29-2019>.
- Duller, G.A.T., 2015. The analyst software package for luminescence data. *Ancient TL* 33, 35–42.
- Durn, G., Čorić, R., Tadej, N., Barudžija, U., Rubinić, V., Husnjak, S., 2014. Bulk and clay mineral composition indicate origin of terra rossa soils in Western Herzegovina. *Geol. Cro.* 67, 171–183. <https://doi.org/10.4154/GC.2014.13>.
- Evans, M.E., Heller, F., 2001. Magnetism of loess/paleosol sequences: recent developments. *Earth Sci. Rev.* 54, 129–144. [https://doi.org/10.1016/S0012-8252\(01\)00044-7](https://doi.org/10.1016/S0012-8252(01)00044-7).
- Fanning, D.S., Keramidas, V.Z., El-Desoky, M.A., 1989. Micas. In: Dixon, J.B., Weed, S.B. (Eds.), *SSSA Book Series. Soil Science Society of America, Madison, WI, USA*, pp. 551–634. <https://doi.org/10.2136/sssabookser1.2ed.c12>.
- Fenn, K., Durcan, J.A., Thomas, D.S.G., Banak, A., 2020. A 180 ka record of environmental change at Erdut (Croatia): a new chronology for the loess–paleosol sequence and its implications for environmental interpretation. *J. Quat. Sci.* 35, 582–593. <https://doi.org/10.1002/jqs.3201>.
- Fitzsimmons, K.E., Marković, S.B., Hambach, U., 2012. Pleistocene environmental dynamics recorded in the loess of the middle and lower Danube basin. *Quat. Sci. Rev.* 41, 104–118. <https://doi.org/10.1016/j.quascirev.2012.03.002>.
- Forster, Th., Evans, M.E., Heller, F., 1994. The frequency dependence of low field susceptibility in loess sediments. *Geophys. J. Int.* 118, 636–642. <https://doi.org/10.1111/j.1365-246X.1994.tb03990.x>.
- Forster, Th., Heller, F., Evans, M.E., Havlíček, P., 1996. Loess in the Czech Republic: magnetic properties and paleoclimate. *Stud. Geophys. Geod.* 40, 243–261. <https://doi.org/10.1007/BF02300741>.
- Frechen, M., Schweitzer, U., Zander, A., 1996. Improvements in sample preparation for the fine grain technique. *Ancient TL* 14, 15–17.
- Frechen, M., Oches, E., Kohfeld, K., 2003. Loess in Europe — Mass accumulation rates during the last Glacial Period. *Quat. Sci. Rev.* 22, 1835–1857. [https://doi.org/10.1016/S0277-3791\(03\)00183-5](https://doi.org/10.1016/S0277-3791(03)00183-5).
- Galović, L., 2014. Geochemical archive in the three loess/paleosol sections in the Eastern Croatia: Zmajevac I, Zmajevac and Erdut. *Aeolian Res.* 15, 113–132. <https://doi.org/10.1016/j.aeolia.2014.07.004>.
- Galović, L., Mileušnić, M., Peh, Z., Durn, G., Halamić, J., 2006. Mineralogical and geochemical characteristics of loess/paleosol section in Šarengrad, Srijem, Croatia. In: Vlahović, I., Tibiljaš, D., Durn, G., Biševac, V. (Eds.), *Abstract Book - 3rd Mid-European Clay Conference – MECC06*, p. 46. Faculty of Science, Faculty of Mining, Geology and Petroleum Engineering, Zagreb.
- Galović, L., Frechen, M., Halamić, J., Durn, G., Romić, M., 2009. Loess chronostratigraphy in Eastern Croatia — A luminescence dating approach. *Quat. Int.* 198, 85–97. <https://doi.org/10.1016/j.quaint.2008.02.004>.
- Gao, X., Hao, Q., Oldfield, F., Bloemendal, J., Deng, C., Wang, L., Song, Y., Ge, J., Wu, H., Li, F., Han, L., Fu, Y., Guo, Z., 2019. New high-temperature dependence of magnetic susceptibility-based climofunction for quantifying paleoprecipitation from Chinese Loess. *Geochim. Geophys. Geosyst.* 20, 4273–4291.
- Guérin, G., Mercier, N., Adamiec, G., 2011. Dose-rate conversion factors: update. *Ancient TL* 29, 5–8.
- Gyalog, L., Síkhegyi, F., 2005. Geological Map of Hungary, M=1:100 000. Geological Institute of Hungary, Budapest.
- Hao, Q., Guo, Z., 2005. Spatial variations of magnetic susceptibility of Chinese loess for the last 600 kyr: implications for monsoon evolution. *J. Geophys. Res.* 110, B12101. <https://doi.org/10.1029/2005JB003765>.
- Hauck, T.C., Lehmkuhl, F., Zeeden, C., Bösen, J., Thiemann, A., Richter, J., 2018. The Aurignacian way of life: contextualizing early modern human adaptation in the Carpathian Basin. *Quat. Int.* 485, 150–166. <https://doi.org/10.1016/j.quaint.2017.10.020>.
- Hećimović, I., 2005. Morphostructural Fabric of Baranja. In: Velić, I., Vlahović, I., Biondić, R. (Eds.), *Abstracts Book. Presented at the 3rd Croatian Geological Congress, Croatian Geological Survey, Zagreb*, pp. 5–6.
- Heller, F., Liu, T., 1982. Magnetostratigraphical dating of loess deposits in China. *Nature* 300, 431–433. <https://doi.org/10.1038/300431a0>.
- Heller, F., Shen, C.D., Beer, J., Liu, X.M., Liu, T.S., Bronger, A., Suter, M., Bonani, G., 1993. Quantitative estimates of pedogenic ferromagnetic mineral formation in Chinese loess and palaeoclimatic implications. *Earth Planet. Sci. Lett.* 114, 385–390. [https://doi.org/10.1016/0012-821X\(93\)90038-B](https://doi.org/10.1016/0012-821X(93)90038-B).
- Hlavatskyi, D.V., Bakhmutov, V.G., 2020. Magnetostratigraphy and magnetic susceptibility of the best developed Pleistocene loess-paleosol sequences of Ukraine: implications for correlation and proposed chronostratigraphic models. *Geol. Q.* 64. <https://doi.org/10.7306/gq.1544>.
- Huntley, D.J., Lamothe, M., 2001. Ubiquity of anomalous fading in K-feldspars and the measurement and correction for it in optical dating. *Can. J. Earth Sci.* 38, 1093–1106. <https://doi.org/10.1139/e01-013>.
- Imbrie, J., Imbrie, J., 1980. Modeling the Climatic Response to Orbital Variations. *Science* 207, 943–953. <https://doi.org/10.1126/science.207.4434.943>.
- Jackson, M.L., 1956. *Soil Chemical Analysis – Advanced Course*. Published by the author, Wisconsin.
- Jackson, M.L., 1957. Frequency distribution of clay minerals in major great soil groups as related to the factors of soil formation. *Clay Clay Miner.* 6, 133–143. <https://doi.org/10.1346/CCMN.1957.0060111>.
- Ji, J., Balsam, W., Chen, J., 2001. Mineralogical and climatic interpretations of the Luochuan Loess Section (China) based on diffuse reflectance spectrophotometry. *Quat. Res.* 56, 23–30. <https://doi.org/10.1006/qres.2001.2238>.
- Ji, J., Balsam, W., Chen, J.U., Liu, L., 2002. Rapid and quantitative measurement of hematite and goethite in the Chinese Loess-paleosol sequence by diffuse reflectance spectroscopy. *Clay Clay Miner.* 50, 208–216. <https://doi.org/10.1346/000986002760832801>.
- Jordanova, D., Hus, J., Geeraerts, R., 2007. Palaeoclimatic implications of the magnetic record from loess/paleosol sequence Viatovo (NE Bulgaria): palaeoclimatic implications of the magnetic record. *Geophys. J. Int.* 171, 1036–1047. <https://doi.org/10.1111/j.1365-246X.2007.03576.x>.
- Kars, R.H., Wallinga, J., Cohen, K.M., 2008. A new approach towards anomalous fading correction for feldspar IRSL dating — Tests on samples in field saturation. *Radiat. Meas.* 43, 786–790. <https://doi.org/10.1016/j.radmeas.2008.01.021>.
- Kovács, J., Raucsik, B., Varga, A., Újvári, G., Varga, G., Ottner, F., 2013. Clay mineralogy of red clay deposits from the central Carpathian Basin (Hungary): implications for Plio-Pleistocene chemical weathering and palaeoclimate. *Turk. J. Earth Sci.* 22, 414–426.
- Kukla, G., 1987. Loess stratigraphy in central China. *Quat. Sci. Rev.* 6, 191–219.
- Laag, C., Hambach, U., Zeeden, C., Lagroix, F., Guyodo, Y., Veres, D., Jovanović, M., Marković, S.B., 2021a. A detailed paleoclimate proxy record for the Middle Danube Basin over the last 430 kyr: a rock magnetic and colorimetric study of the Zemun loess-paleosol sequence. *Front. Earth Sci.* <https://doi.org/10.3389/feart.2021.600086>, 9 (May).
- Laag, C., Jordanova, D., Lagroix, F., Jordanova, N., Guyodo, Y., 2021b. A New Reference Loess-Paleosol Archive Spanning the Last 850 Kyr near Plevna (Bulgaria): First Results. <https://doi.org/10.5194/egusphere-egu21-6381>. EGU 2021, Vienna.
- Lagroix, F., Guyodo, Y., 2017. A New tool for separating the magnetic mineralogy of complex mineral assemblages from low temperature magnetic behavior. *Front. Earth Sci.* 5, 61. <https://doi.org/10.3389/feart.2017.00061>.
- Lamothe, M., Auclair, M., Hamzaoui, C., Huot, S., 2003. Towards a prediction of long-term anomalous fading of feldspar IRSL. *Radiat. Meas.* 37, 493–498. [https://doi.org/10.1016/S1350-4487\(03\)00016-7](https://doi.org/10.1016/S1350-4487(03)00016-7).
- Li, Y., Tsukamoto, S., Frechen, M., Gabriel, G., 2018. Timing of fluvial sedimentation in the Upper Rhine Graben since the Middle Pleistocene: constraints from quartz and feldspar luminescence dating. *Boreas* 47, 256–270. <https://doi.org/10.1111/bor.12266>.
- Lisiecki, L.E., Raymo, M.E., 2005. A Pliocene-Pleistocene stack of 57 globally distributed benthic $\delta^{18}O$ records. *Paleoceanography* 20. <https://doi.org/10.1029/2004PA001071>.
- Liu, X., Rolph, T., Bloemendal, J., Shaw, J., Liu, T., 1995. Quantitative estimates of paleoprecipitation at Xifeng, in the Loess Plateau of China. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 113, 243–248. [https://doi.org/10.1016/0013-0182\(95\)00053-0](https://doi.org/10.1016/0013-0182(95)00053-0).
- Liu, Q., Roberts, A.P., Larrasoana, J.C., Banerjee, S.K., Guyodo, Y., Tauxe, L., Oldfield, F., 2012. Environmental magnetism: principles and applications. *Rev. Geophys.* 50, RG4002. <https://doi.org/10.1029/2012RG000393>.
- Lukić, T., Basarin, B., Bugle, B., Marković, S.B., Tomović, V.M., Raljić, J.P., Hrnjak, I., Timar-Gabor, A., Hambach, U., Gavrilov, M.B., 2014. A joined rock magnetic and colorimetric perspective on the Late Pleistocene climate of Orlovat loess site (Northern Serbia). *Quat. Int.* 334–335, 179–188. <https://doi.org/10.1016/j.quaint.2014.03.042>.
- Maher, B.A., 1986. Characterisation of soils by mineral magnetic measurements. *Phys. Earth Planet. Inter.* 42, 76–92. [https://doi.org/10.1016/S0031-9201\(86\)80010-3](https://doi.org/10.1016/S0031-9201(86)80010-3).
- Maher, B.A., 1998. Magnetic properties of modern soils and Quaternary loessic paleosols: paleoclimatic implications. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 137, 25–54.
- Maher, B.A., 2011. The magnetic properties of Quaternary aeolian dusts and sediments, and their palaeoclimatic significance. *Aeolian Res.* 3, 87–144. <https://doi.org/10.1016/j.aeolia.2011.01.005>.
- Maher, B.A., Thompson, R., Zhou, L.P., 1994. Spatial and temporal reconstructions of changes in the Asian palaeomonsoon: a new mineral magnetic approach. *Earth Planet. Sci. Lett.* 125, 461–471. [https://doi.org/10.1016/0012-821X\(94\)90232-1](https://doi.org/10.1016/0012-821X(94)90232-1).
- Maher, B.A., Alekseev, A., Alekseeva, T., 2003. Magnetic mineralogy of soils across the Russian Steppe: climatic dependence of pedogenic magnetite formation. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 201, 321–341. [https://doi.org/10.1016/S0031-0182\(03\)00618-7](https://doi.org/10.1016/S0031-0182(03)00618-7).
- Marković, S., Kostić, N.S., Oches, E.A., 2004. Paleosols in the Ruma loess section (Vojvodina, Serbia). *Revi. Mex. Ciencias Geol.* 21, 79–87.
- Marković, S.B., Hambach, U., Catto, N., Jovanović, M., Bugle, B., Machalet, B., Zöller, L., Glaser, B., Frechen, M., 2009. Middle and Late Pleistocene loess sequences at Batanjica, Vojvodina, Serbia. *Quat. Int.* 198, 255–266. <https://doi.org/10.1016/j.quaint.2008.12.004>.
- Marković, S.B., Hambach, U., Stevens, T., Kukla, G.J., Heller, F., McCoy, W.D., Oches, E. A., Bugle, B., Zöller, L., 2011. The last million years recorded at the Stari Slankamen (Northern Serbia) loess-paleosol sequence: revised chronostratigraphy and long-term environmental trends. *Quat. Sci. Rev.* 30, 1142–1154. <https://doi.org/10.1016/j.quascirev.2011.02.004>.
- Marković, S.B., Hambach, U., Stevens, T., Jovanović, M., O'Hara-Dhand, K., Basarin, B., Lu, H., Smalley, I., Bugle, B., Zech, M., Svirčev, Z., Sümege, P., Milojković, N., Zöller, L., 2013. Loess in the Vojvodina region (Northern Serbia): an essential link between European and Asian Pleistocene environments. *Neth. J. Geosci.* 91, 173–188. <https://doi.org/10.1017/S0016774600001578>.

- Marković, S.B., Stevens, T., Kukla, G.J., Hambach, U., Fitzsimmons, K.E., Gibbard, P., Buggle, B., Zech, M., Guo, Z., Hao, Q., Wu, H., O'Hara Dhand, K., Smalley, I.J., Újvári, G., Sümegi, P., Timar-Gabor, A., Veres, D., Sirocko, F., Vasiljević, D.A., Jary, Z., Svensson, A., Jović, V., Lehmkuhl, F., Kovács, J., Svirčev, Z., 2015. Danube loess stratigraphy — Towards a pan-European loess stratigraphic model. *Earth Sci. Rev.* 148, 228–258. <https://doi.org/10.1016/j.earscirev.2015.06.005>.
- Meyers, S.R., 2014. *astrochron: An R Package for Astrochronology Version 0.8*. Millot, G., 1964. *Géologie des argiles*. Masson and Cie, Paris.
- Moore, D.M., Reynolds Jr., R.C., 1997. *X-Ray Diffraction and the Identification and Analysis of Clay Minerals*. Oxford University Press, Oxford.
- Murray, A.S., Schmidt, E.D., Stevens, T., Buylaert, J.-P., Marković, S.B., Tsukamoto, S., Frechen, M., 2014. Dating Middle Pleistocene loess from Stari Slankamen (Vojvodina, Serbia) — Limitations imposed by the saturation behaviour of an elevated temperature IRSL signal. *CATENA* 117, 34–42. <https://doi.org/10.1016/j.catena.2013.06.029>.
- Novothy, A., Frechen, M., Horváth, E., Wacha, L., Rolf, C., 2011. Investigating the penultimate and last glacial cycles of the Süttő loess section (Hungary) using luminescence dating, high-resolution grain size, and magnetic susceptibility data. *Quat. Int.* 234, 75–85. <https://doi.org/10.1016/j.quaint.2010.08.002>.
- Novothy, A., Barta, G., Végh, T., Bradák, B., Surányi, G., Horváth, E., 2020. Correlation of drilling cores and the Paks brickyard jet section at the area of Paks, Hungary. *Quat. Int.* 552, 50–61. <https://doi.org/10.1016/j.quaint.2019.09.012>.
- Obrecht, I., Buggle, B., Catto, N., Marković, S.B., Bösel, S., Vandenberghe, D.A.G., Hambach, U., Svirčev, Z., Lehmkuhl, F., Basarin, B., Gavrilov, M.B., Jović, G., 2014. The Late Pleistocene Belotina section (southern Serbia) at the southern limit of the European loess belt: Environmental and climate reconstruction using grain size and stable C and N isotopes. *Quat. Int.* 334–335, 10–19. <https://doi.org/10.1016/j.quaint.2013.05.037>.
- Obrecht, I., Zeeden, C., Hambach, U., Veres, D., Marković, S.B., Bösen, J., Svirčev, Z., Bačević, N., Gavrilov, M.B., Lehmkuhl, F., 2016. Tracing the influence of Mediterranean climate on Southeastern Europe during the past 350,000 years. *Sci. Rep.* 6, 36334. <https://doi.org/10.1038/srep36334>.
- Obrecht, I., Zeeden, C., Hambach, U., Veres, D., Marković, S.B., Lehmkuhl, F., 2019. A critical reevaluation of palaeoclimate proxy records from loess in the Carpathian Basin. *Earth Sci. Rev.* 190, 498–520. <https://doi.org/10.1016/j.earscirev.2019.01.020>.
- PAGES, Past Interglacials Working Group, 2016. Interglacials of the last 800,000 years. *Rev. Geophys.* 54, 162–219. <https://doi.org/10.1002/2015RG000482>.
- Pécsi, M., 1990. Loess is not just the accumulation of dust. *Quat. Int.* 7–8, 1–21. [https://doi.org/10.1016/1040-6182\(90\)90034-2](https://doi.org/10.1016/1040-6182(90)90034-2).
- Pécsi, M., 1993. Quaternary and loess research. In: Bassa, L., Keresztesi, Zs, Lóczy, D. (Eds.), *Presented at the Loess in Form 2*. Hungarian Academy of Science, Budapest, pp. 1–82.
- Peters, C., Dekkers, M.J., 2003. Selected room temperature magnetic parameters as a function of mineralogy, concentration and grain size. *Phys. Chem. Earth Parts A/B/C* 28, 659–667. [https://doi.org/10.1016/S1474-7065\(03\)00120-7](https://doi.org/10.1016/S1474-7065(03)00120-7).
- Pikija, M., Šikić, K., Trifunović, S., 1991a. Basic Geological Map 1:100 000, Sheet Mohač L 34-7474 (2015).
- Pikija, M., Šikić, K., Trifunović, S., 1991b. Basic Geological Map, Explanatory notes, Sheet Mohač L 34-74.
- Pikija, M., Šikić, K., Sarkotić-Šlat, M., Magaš, N., 1995. *Geologija hrvatskog dijela Baranje*. In: Vlahović, I., Velić, I., Šparica, M. (Eds.), *First Croatian Geological Congress*. Presented at the First Croatian Geological Congress, Opatija, October 18–21, Proceedings, 2. Institute of Geology & Croatian Geological Society, Zagreb, Opatija, pp. 447–451.
- Prescott, J.R., Hutton, J.T., 1994. Cosmic ray contributions to dose rates for luminescence and ESR dating: large depths and long-term time variations. *Radiat. Meas.* 23, 497–500. [https://doi.org/10.1016/1350-4487\(94\)90086-8](https://doi.org/10.1016/1350-4487(94)90086-8).
- Prescott, J.R., Stephan, L.G., 1982. The contribution of cosmic radiation to the environmental dose for thermoluminescent dating — latitude, altitude and depth dependences. *PACT* 6, 17–25.
- Pye, K., 1987. *Aeolian Dust and Dust Deposits*. Academic Press, London.
- Rees-Jones, J., 1995. Optical dating of young sediments using fine-grain quartz. *Ancient TL* 9–14.
- Rubinić, V., Galović, L., Lazarević, B., Husnjak, S., Durn, G., 2018. Pseudogleyed loess derivatives — the most common soil parent materials in the Pannonian region of Croatia. *Quat. Int.* 494, 248–262. <https://doi.org/10.1016/j.quaint.2017.06.044>.
- Schaetzl, R.J., Bettis, E.A., Crouvi, O., Fitzsimmons, K.E., Grimley, D.A., Hambach, U., Lehmkuhl, F., Marković, S.B., Mason, J.A., Owczarek, P., Roberts, H.M., Rousseau, D.-D., Stevens, T., Vandenberghe, J., Zárate, M., Veres, D., Yang, S., Zech, M., Conroy, J.L., Dave, A.K., Faust, D., Hao, Q., Obrecht, I., Prud'homme, C., Smalley, I., Tripaldi, A., Zeeden, C., Zech, R., 2018. Approaches and challenges to the study of loess—Introduction to the LoessFest special issue. *Quat. Res.* 89, 563–618. <https://doi.org/10.1017/qua.2018.15>.
- Schatz, A.-K., Buylaert, J.-P., Murray, A., Stevens, T., Scholten, T., 2012. Establishing a luminescence chronology for a palaeosol-loess profile at Tokaj (Hungary): a comparison of quartz OSL and polymineral IRSL signals. *Quat. Geochronol.* 10, 68–74. <https://doi.org/10.1016/j.quageo.2012.02.018>.
- Schmidt, E.D., Machalet, B., Marković, S.B., Tsukamoto, S., Frechen, M., 2010. Luminescence chronology of the upper part of the Stari Slankamen loess sequence (Vojvodina, Serbia). *Quat. Geochronol.* 5, 137–142. <https://doi.org/10.1016/j.quageo.2009.09.006>.
- Schmidt, E.D., Tsukamoto, S., Frechen, M., Murray, A.S., 2014. Elevated temperature IRSL dating of loess sections in the East Eifel region of Germany. *Quat. Int.* 334–335, 141–154. <https://doi.org/10.1016/j.quaint.2014.03.006>.
- Schulte, P., Lehmkuhl, F., 2018. The difference of two laser diffraction patterns as an indicator for post-depositional grain size reduction in loess-paleosol sequences. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 509, 126–136. <https://doi.org/10.1016/j.palaeo.2017.02.022>.
- Sheldon, N.D., Tabor, N.J., 2009. Quantitative paleoenvironmental and paleoclimatic reconstruction using paleosols. *Earth Sci. Rev.* 95, 1–52. <https://doi.org/10.1016/j.earscirev.2009.03.004>.
- Smalley, I., O'Hara-Dhand, K., Wint, J., Machalet, B., Jary, Z., Jefferson, I., 2009. Rivers and loess: the significance of long river transportation in the complex event-sequence approach to loess deposit formation. *Quat. Int.* 198, 7–18. <https://doi.org/10.1016/j.quaint.2008.06.009>.
- Sommerwerk, N., Hein, T., Schneider-Jacoby, M., Baumgartner, C., Ostojic, A., Siber, R., Bloesch, J., Pannonić, M., Tockner, K., 2009. The Danube River Basin. In: *Rivers of Europe*. Elsevier, pp. 59–112. <https://doi.org/10.1016/B978-0-12-369449-2.00003-5>.
- Song, Y., Guo, Z., Marković, S., Hambach, U., Deng, C., Chang, L., Wu, J., Hao, Q., 2018. Magnetic stratigraphy of the Danube loess: a composite Titel-Stari Slankamen loess section over the last one million years in Vojvodina, Serbia. *J. Asian Earth Sci.* 155, 68–80. <https://doi.org/10.1016/j.jseas.2017.11.012>.
- Spooner, N.A., 1994. The anomalous fading of infrared-stimulated luminescence from feldspars. *Radiat. Meas.* 23, 625–632. [https://doi.org/10.1016/1350-4487\(94\)90111-2](https://doi.org/10.1016/1350-4487(94)90111-2).
- Starkey, H.C., Blackmon, P.D., Hauff, P.L., 1984. The routine mineralogical analysis of clay-bearing samples (No. 1563). In: *U. S. Geological Survey Bulletin*. <https://doi.org/10.3133/b1563>.
- Sümegi, P., Gulyás, S., Molnár, D., Sümegi, B.P., Almond, P.C., Vandenberghe, J., Zhou, L., Pál-Molnár, E., Töröcsik, T., Hao, Q., Smalley, I., Molnár, M., Marsi, I., 2018. New chronology of the best developed loess/paleosol sequence of Hungary capturing the past 1.1 Ma: implications for correlation and proposed pan-Eurasian stratigraphic schemes. *Quat. Sci. Rev.* 191, 144–166.
- Taheri, M., Khormali, F., Wang, X., Amini, A., Landi, A., Wei, H., Kehl, M., Chen, F., 2020. Clay mineralogy and geochemistry of the Lower Pleistocene Loess in the Iranian Loess Plateau (Agh Band section) and implications for its provenance and paleoclimate change. *Quat. Int.* 552, 91–99. <https://doi.org/10.1016/j.quaint.2019.09.011>.
- Thamó-Boszó, E., Kovács, L.Ó., Magyari, Á., Marsi, I., 2014. Tracing the origin of loess in Hungary with the help of heavy mineral composition data. *Quat. Int.* 319, 11–21. <https://doi.org/10.1016/j.quaint.2013.04.030>.
- Thiel, C., Buylaert, J.-P., Murray, A., Terhorst, B., Hofer, I., Tsukamoto, S., Frechen, M., 2011. Luminescence dating of the Stratzing loess profile (Austria) — Testing the potential of an elevated temperature post-IR IRSL protocol. *Quat. Int.* 234, 23–31. <https://doi.org/10.1016/j.quaint.2010.05.018>.
- Thiel, C., Horváth, E., Frechen, M., 2014. Revisiting the loess/paleosol sequence in Paks, Hungary: A post-IR IRSL based chronology for the 'Young Loess Series'. *Quat. Int.* 319, 88–98. <https://doi.org/10.1016/j.quaint.2013.05.045>.
- Thompson, R., Oldfield, F., 1986. *Magnetic Properties of Natural Materials*, in: *Environmental Magnetism*. Springer Netherlands, Dordrecht, pp. 21–38. https://doi.org/10.1007/978-94-011-8036-8_4.
- Trifunović, S., Strajin, V., 1995. *Basic Geological Map of Serbia, 1:100 000, Sheet Mohač*. Savezno ministarstvo za privrednu SRJ, Beograd.
- Újvári, G., Varga, A., Ramos, F.C., Kovács, J., Németh, T., Stevens, T., 2012. Evaluating the use of clay mineralogy, Sr–Nd isotopes and zircon U–Pb ages in tracking dust provenance: an example from loess of the Carpathian Basin. *Chem. Geol.* 304–305, 83–96. <https://doi.org/10.1016/j.chemgeo.2012.02.007>.
- Újvári, G., Molnár, M., Novothy, Á., Pál-Gergely, B., Kovács, J., Várhegyi, A., 2014a. AMS 14C and OSL/IRSL dating of the Dunaszekcső loess sequence (Hungary): chronology for 20 to 150 ka and implications for establishing reliable age–depth models for the last 40 ka. *Quat. Sci. Rev.* 106, 140–154. <https://doi.org/10.1016/j.quascirev.2014.06.009>.
- Újvári, G., Varga, A., Raucsik, B., Kovács, J., 2014b. The Paks loess-paleosol sequence: a record of chemical weathering and provenance for the last 800ka in the mid-Carpathian Basin. *Quat. Int.* 319, 22–37. <https://doi.org/10.1016/j.quaint.2012.04.004>.
- Újvári, G., Kok, J.F., Varga, G., Kovács, J., 2016a. The physics of wind-blown loess: implications for grain size proxy interpretations in Quaternary paleoclimate studies. *Earth Sci. Rev.* 154, 247–278. <https://doi.org/10.1016/j.earscirev.2016.01.006>.
- Újvári, G., Molnár, M., Pál-Gergely, B., 2016b. Charcoal and mollusc shell 14 C-dating of the Dunaszekcső loess record, Hungary. *Quat. Geochronol.* 35, 43–53. <https://doi.org/10.1016/j.quageo.2016.05.005>.
- Újvári, G., Stevens, T., Molnár, M., Demény, A., Lambert, F., Varga, G., Jull, A.J.T., Pál-Gergely, B., Buylaert, J.-P., Kovács, J., 2017. Coupled European and Greenland last glacial dust activity driven by North Atlantic climate. *Proc. Natl. Acad. Sci. U. S. A.* 114, E10632–E10638. <https://doi.org/10.1073/pnas.1712651114>.
- Vandenberghe, J., 2013. Grain size of fine-grained windblown sediment: a powerful proxy for process identification. *Earth Sci. Rev.* 121, 18–30. <https://doi.org/10.1016/j.earscirev.2013.03.001>.
- Vandenberghe, J.F., Múcher, H.J., Roebroeks, W., Gemke, D., 1985. *Lithostratigraphy and palaeoenvironment of the Pleistocene deposits at Maastricht-Belvédère, southern Limburg, The Netherlands*. *Meded. Rijks Geol. Dienst* 39, 7–29.
- Vandenberghe, J., Marković, S.B., Jovanović, M., Hambach, U., 2014. Site-specific variability of loess and palaeosols (Ruma, Vojvodina, northern Serbia). *Quat. Int.* 334–335, 86–93. <https://doi.org/10.1016/j.quaint.2013.10.036>.
- Vandenberghe, J., Sun, Y., Wang, X., Abels, H.A., Liu, X., 2018. Grain-size characterization of reworked fine-grained aeolian deposits. *Earth Sci. Rev.* 177, 43–52. <https://doi.org/10.1016/j.earscirev.2017.11.005>.

- Varga, A., Újvári, G., Raucsik, B., 2011. Tectonic versus climatic control on the evolution of a loess–paleosol sequence at Beremend, Hungary: an integrated approach based on paleoecological, clay mineralogical, and geochemical data. *Quat. Int.* 240, 71–86. <https://doi.org/10.1016/j.quaint.2010.10.032>.
- Wacha, L., Frechen, M., 2011. The geochronology of the “Gorjanović loess section” in Vukovar, Croatia. *Quat. Int.* 240, 87–99. <https://doi.org/10.1016/j.quaint.2011.04.010>.
- Wacha, L., Galović, L., Koloszar, L., Magyari, Á., Chikán, G., Marsi, I., 2013. The chronology of the Šarengrad II loess-paleosol section (Eastern Croatia). *Geol. Cro.* 66, 191–203. <https://doi.org/10.4154/GC.2013.18>.
- Wacha, L., Vlahović, I., Tsukamoto, S., Kovačić, M., Hasan, O., Pavelić, D., 2016. The chronostratigraphy of the latest Middle Pleistocene aeolian and alluvial activity on the Island of Hvar, eastern Adriatic, Croatia. *Boreas* 45, 152–164. <https://doi.org/10.1111/bor.12141>.
- Wacha, L., Matoš, B., Kunz, A., Lužar-Oberiter, B., Tomljenović, B., Banak, A., 2018a. First post-IR IRSL dating results of Quaternary deposits from Bilogora (NE Croatia): implications for the Pleistocene relative uplift and incision rates in the area. *Quat. Int.* 494, 193–210. <https://doi.org/10.1016/j.quaint.2017.08.049>.
- Wacha, L., Rolf, C., Hambach, U., Frechen, M., Galović, L., Duchoslav, M., 2018b. The Last Glacial aeolian record of the Island of Susak (Croatia) as seen from a high-resolution grain-size and rock magnetic analysis. *Quat. Int.* 494, 211–224. <https://doi.org/10.1016/j.quaint.2017.08.016>.
- Wacha, L., Laag, C., Grizelj, A., Tsukamoto, S., Zeeden, C., Ivanišević, D., Rolf, C., Banak, A., Frechen, M., 2020. Zmajevac data. <https://doi.org/10.17632/RNVRZFXGFM.1>.
- Weaver, C.E., 1989. *Clays, Muds, and Shales*. Elsevier, Amsterdam.
- Wilson, M.J., 1999. The origin and formation of clay minerals in soils; past, present and future perspectives. *Clay Miner.* 34, 7–25.
- Wintle, A.G., 1973. Anomalous Fading of Thermo-luminescence in Mineral Samples. *Nature* 245, 143–144. <https://doi.org/10.1038/245143a0>.
- Wu, Y., Qiu, S., Fu, S., Rao, Z., Zhu, Z., 2018. Pleistocene climate change inferred from multi-proxy analyses of a loess-paleosol sequence in China. *J. Asian Earth Sci.* 154, 428–434. <https://doi.org/10.1016/j.jseas.2017.10.007>.
- Zeeden, C., Kels, H., Hambach, U., Schulte, P., Protze, J., Eckmeier, E., Marković, S.B., Klasen, N., Lehmkuhl, F., 2016. Three climatic cycles recorded in a loess-paleosol sequence at Semlac (Romania) – Implications for dust accumulation in south-eastern Europe. *Quat. Sci. Rev.* 154, 130–142. <https://doi.org/10.1016/j.quascirev.2016.11.002>.